

# **WHEN GOVERNMENTS SAY ‘GREEN’ — DO BOND MARKETS BELIEVE THEM?**

By

Nahian Ibnat

Submitted to

Central European University — Private University

Department of Economics

*In partial fulfilment of the requirements for the degree of  
Master of Arts in Economics, Data and Policy*

Supervisor: Zoltán Csaba Tóth

Vienna, Austria

2026

# Copyright Notice

Copyright © Nahian Ibnat, 2026. *When Governments Say ‘Green’ — Do Bond Markets Believe Them?* — This work is licensed under [Creative Commons Attribution \(CC BY\) 4.0 International](https://creativecommons.org/licenses/by/4.0/) license.



For bibliographic and reference purposes this thesis should be referred to as:

Ibnat, N. 2026. *When Governments Say ‘Green’ — Do Bond Markets Believe Them?* MA Thesis, Department of Economics, Central European University — Private University, Vienna.

---

<sup>1</sup>Icon by Creative Commons: <https://creativecommons.org/mission/downloads/>

## Author's Declaration

I, Nahian Ibnat, candidate for the MA degree in Economics, Data and Policy, declare herewith that the present thesis titled “*When Governments Say ‘Green’ — Do Bond Markets Believe Them?*” is exclusively my own work, based on my research and only such external information as properly credited in notes and bibliography. I declare that no unidentified and illegitimate use was made of the work of others, and no part of the thesis infringes any person's or institution's copyright.

I also declare that no part of the thesis has been submitted in this form to any other institution of higher education for an academic degree.

Vienna, 08 June 2026



---

Signature

## Declaration of Generative AI

Generative artificial intelligence (GenAI) was used in this work. I, Nahian Ibnat, have reviewed and edited the content as needed and take full responsibility for the content, claims, and references. An overview of the use of GenAI is provided below.

- I used Claude (Anthropic, <https://claude.ai>) to support the drafting, paraphrasing, and structural editing of thesis chapters, including the introduction, literature review, methodology, results, discussion, and conclusion. The tool was used to improve sentence clarity, suggest paragraph structure, and check for coherence and consistency across sections.
- I used Claude (Anthropic, <https://claude.ai>) to assist with the development, debugging, and documentation of Python notebooks for data processing, Greenium construction, regression estimation (OLS, HC3, PCSE via linearmodels), stacked Difference-in-Differences design, event study construction, parallel trends testing, placebo tests, and figure generation using matplotlib. The tool was used to identify syntax errors, suggest code improvements, and generate documentation strings.
- The complete replication code, data pipeline, and Jupyter notebooks for this thesis are available at: <https://github.com/NahianIbnat-2000/Thesis>

# Abstract

This thesis tests whether political credibility signals shift the sovereign green bond Greenium — the yield concession investors accept to hold a green-labelled sovereign bond relative to a comparable conventional bond in emerging market economies. Using a twin-bond stacked Difference-in-Differences design applied to India and Indonesia, the thesis tests three sequential hypotheses: whether daily GDELT V2Tone climate sentiment predicts Greenium changes, whether signalling noise from contradictory government messaging compresses the Greenium, and whether inaugural sovereign green bond issuance or a major internationally backed climate commitment produces a structural level shift. All three hypotheses produce results that are not statistically different from zero. A European robustness check recovers persistent greeniums of  $-1.82$  and  $-2.80$  basis points in Germany and Denmark, confirming the methodology detects greeniums where they exist. The findings are attributed to a market development threshold: below a certain level of institutional investor depth and secondary market liquidity, credibility signals cannot reach daily bond prices regardless of their genuine content.

**Keywords:** sovereign green bonds, greenium, difference-in-differences, political credibility, emerging markets, GDELT, climate finance

## Acknowledgements

I would like to express my sincere gratitude to my supervisor, Professor Zoltán Csaba Tóth, for his guidance and support throughout the development of this thesis. Despite this being his first experience supervising a Master's thesis, he was consistently responsive, engaged, and generous with his suggestions at every stage of the research. His expertise, patience, and encouragement were instrumental in shaping this work.

Special thanks are due to Professor Tomy Lee for his constructive feedback during the Thesis Seminar presentations. His comments helped sharpen the empirical framing and pushed the analysis in more rigorous directions. I am also grateful to my Academic Writing Instructor, Zsuzsanna Tóth, whose writing instruction improved both the clarity and coherence of the thesis considerably.

I am grateful to the CEU Library for providing access to Refinitiv Eikon through the university's institutional subscription. The entire empirical pipeline of this thesis depends on the bond yield data collected through that platform, and this research would not have been possible without it.

My sincere thanks go to the Department Coordinators Anita Apor, Dominika Dash, and Judit Lafferthon, who were unfailingly helpful with every administrative and academic matter, large or small, throughout the programme.

Finally, I owe a great deal to my dear friends Boga Petruska and Yelyzaveta Bendih, who supported me through proofreading, technical troubleshooting, and the day-to-day challenges of thesis writing. Their encouragement made the entire journey both more manageable and more enjoyable.

# Contents

|                                                                           |            |
|---------------------------------------------------------------------------|------------|
| <b>Copyright Notice</b>                                                   | <b>i</b>   |
| <b>Author’s Declaration</b>                                               | <b>ii</b>  |
| <b>Declaration of Generative AI</b>                                       | <b>iii</b> |
| <b>Abstract</b>                                                           | <b>iv</b>  |
| <b>Acknowledgements</b>                                                   | <b>v</b>   |
| <b>List of Abbreviations</b>                                              | <b>xii</b> |
| <b>1 Introduction</b>                                                     | <b>1</b>   |
| <b>2 Literature Review</b>                                                | <b>4</b>   |
| 2.1 Core Concepts and Definitions . . . . .                               | 4          |
| 2.2 Green Bond Pricing: What the Evidence Shows . . . . .                 | 5          |
| 2.3 Theoretical Frameworks . . . . .                                      | 6          |
| 2.3.1 Credible Commitment Theory . . . . .                                | 6          |
| 2.3.2 Asymmetric Information and Signalling . . . . .                     | 7          |
| 2.4 Climate Sentiment and NLP in Financial Markets . . . . .              | 7          |
| 2.5 Sovereign Green Bonds in Emerging Market Economies: The Gap . . . . . | 8          |
| 2.6 Identification Strategy . . . . .                                     | 8          |
| 2.7 Research Questions . . . . .                                          | 9          |
| <b>3 Data and Methodology</b>                                             | <b>11</b>  |
| 3.1 Sample Selection and Twin-Bond Design . . . . .                       | 11         |
| 3.2 The Greenium . . . . .                                                | 12         |
| 3.3 Control Variables . . . . .                                           | 12         |
| 3.4 Identification Strategy . . . . .                                     | 13         |
| 3.4.1 First-Differencing . . . . .                                        | 13         |
| 3.4.2 The Full Model Sequence: M1 through M5 . . . . .                    | 13         |
| 3.4.3 The Stacked DiD . . . . .                                           | 15         |

|          |                                                                        |           |
|----------|------------------------------------------------------------------------|-----------|
| 3.5      | Parallel Trends Validation . . . . .                                   | 16        |
| 3.6      | Robustness Checks . . . . .                                            | 17        |
| 3.6.1    | Standard Error Robustness . . . . .                                    | 17        |
| 3.6.2    | Placebo Tests . . . . .                                                | 17        |
| 3.6.3    | European Extension . . . . .                                           | 17        |
| 3.6.4    | Granger Causality . . . . .                                            | 17        |
| <b>4</b> | <b>Results: Asia (India and Indonesia)</b>                             | <b>18</b> |
| 4.1      | What the Data Look Like Before Any Regression . . . . .                | 18        |
| 4.2      | Parallel Trends: The Identification Holds for India . . . . .          | 19        |
| 4.3      | Main Regression Results . . . . .                                      | 19        |
| 4.4      | Stacked DiD: Confirming the Null with Cleaner Identification . . . . . | 21        |
| 4.5      | Robustness . . . . .                                                   | 21        |
| 4.6      | Economic Interpretation: What the Null Means . . . . .                 | 22        |
| <b>5</b> | <b>Robustness: European Extension (Germany and Denmark)</b>            | <b>24</b> |
| 5.1      | The European Greenium Exists and Is Stable . . . . .                   | 24        |
| 5.2      | The DiD Finds No Post-Issuance Shift — and That Is Expected . . . . .  | 25        |
| 5.3      | Post-Issuance Dynamics: No Systematic Pattern . . . . .                | 26        |
| 5.4      | What the European Results Establish . . . . .                          | 26        |
| <b>6</b> | <b>Discussion and Comparative Analysis</b>                             | <b>27</b> |
| 6.1      | The Same Regression Output, Two Different Stories . . . . .            | 27        |
| 6.2      | Why the Information Channels Are Silent . . . . .                      | 29        |
| 6.3      | A Threshold Interpretation . . . . .                                   | 29        |
| 6.4      | What Moves the Greenium in Asia, and Why It Matters . . . . .          | 30        |
| 6.5      | Policy Implications . . . . .                                          | 30        |
| 6.6      | Limitations . . . . .                                                  | 31        |
| <b>7</b> | <b>Conclusion</b>                                                      | <b>33</b> |
| <b>A</b> | <b>Bond Pair ISINs and Data Sources</b>                                | <b>39</b> |
| <b>B</b> | <b>Descriptive Statistics</b>                                          | <b>40</b> |
| <b>C</b> | <b>Full Regression Tables</b>                                          | <b>42</b> |
| <b>D</b> | <b>Standard Error Robustness</b>                                       | <b>43</b> |
| <b>E</b> | <b>Placebo Tests</b>                                                   | <b>44</b> |
| <b>F</b> | <b>Parallel Trends Formal F-Tests</b>                                  | <b>45</b> |



|          |                                    |           |
|----------|------------------------------------|-----------|
| <b>G</b> | <b>European Robustness Results</b> | <b>47</b> |
| <b>H</b> | <b>Stacked DiD Verification</b>    | <b>50</b> |
| <b>I</b> | <b>Granger Causality Tests</b>     | <b>51</b> |

# List of Tables

|     |                                                                   |    |
|-----|-------------------------------------------------------------------|----|
| 3.1 | Model Sequence Summary . . . . .                                  | 15 |
| 4.1 | Main Regression Results — Asia Panel . . . . .                    | 20 |
| 4.2 | Stacked DiD Results — Asia . . . . .                              | 21 |
| 4.3 | Standard Error Robustness — Asia . . . . .                        | 22 |
| 5.1 | European Results — Descriptive and DiD . . . . .                  | 25 |
| 6.1 | Four-Country Comparison . . . . .                                 | 28 |
| A.1 | Bond Pair ISINs and Data Sources . . . . .                        | 39 |
| B.1 | Descriptive Statistics — Asia Panel . . . . .                     | 40 |
| C.1 | Full Regression Results — Asia Panel . . . . .                    | 42 |
| D.1 | Standard Error Robustness — Asia . . . . .                        | 43 |
| E.1 | Placebo Tests — Asia (M5 Pooled Specification) . . . . .          | 44 |
| F.1 | Parallel Trends Joint F-Tests — Asia . . . . .                    | 45 |
| G.1 | European Robustness Regression Results . . . . .                  | 47 |
| G.2 | Post-Issuance Joint F-Tests — Europe . . . . .                    | 47 |
| I.1 | Granger Causality Tests — Signalling Noise and Greenium . . . . . | 52 |

# List of Figures

|     |                                                                                                                                                                                                                                                                                                                                     |    |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| B.1 | Greenium by twin pair — 10Y sovereign bonds, India and Indonesia (30-day moving average). Vertical dashed lines mark the JETP G20 (15 Nov 2022) and India 2nd auction (27 Oct 2023) treatment dates. Source: Refinitiv Eikon; author’s calculations. . . . .                                                                        | 41 |
| B.2 | Daily GDELT V2Tone climate sentiment — India and Indonesia, 2020–2024. The 30-day moving average (bold line) is flat throughout with no structural break at either treatment date, consistent with the H1 null result. Treatment dates marked by vertical dashed lines. . . . .                                                     | 41 |
| F.1 | Event study — India (treatment: 2nd green bond auction, 27 Oct 2023). Pre-event window: Jan–Oct 2023 (~120 trading days). Pre-event coefficients are jointly flat — $F(9, 84) = 1.006$ , $p = 0.44$ — validating the DiD identification. No post-event shift detected (H3 null). . . . .                                            | 45 |
| F.2 | Event study — Indonesia (treatment: JETP G20 Bali, 15 Nov 2022). Pre-event window limited to ~55 trading days from bond start. Weeks –10 and –9 are structural zeros (bond did not yet exist). Formal parallel trends test passes — $F(7, 75) = 0.740$ , $p = 0.64$ — but with limited statistical power. . . . .                   | 46 |
| G.1 | European sovereign green bond greenium (30-day moving average) — Germany and Denmark. Both series show persistently negative, stable greeniums throughout the full sample. Vertical dashed lines mark inaugural issuance dates (Germany: 2 Sep 2020; Denmark: 21 Jan 2022). Source: Refinitiv Eikon; author’s calculations. . . . . | 48 |
| G.2 | Post-issuance greenium dynamics — Germany and Denmark (event study, side-by-side). Both countries show minor fluctuations around zero with no systematic post-issuance trend, consistent with a stable equilibrium greenium established at inaugural issuance. . . . .                                                              | 48 |
| G.3 | Germany — post-issuance price discovery (inaugural Green Bund, 2 Sep 2020, 20-week window). Pre-issuance region (grey) does not test parallel trends. Post-issuance coefficients are jointly non-significant — $F(20, 84) = 0.603$ , $p = 0.900$ . . . . .                                                                          | 49 |

|     |                                                                                                                                                                                                                                         |    |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| G.4 | Denmark — post-issuance greenium dynamics (inaugural sovereign green bond, 21 Jan 2022, 20-week window). Pre-issuance region is thin. Post-issuance coefficients are jointly non-significant — $F(20, 87) = 0.100, p = 1.000$ . . . . . | 49 |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|

## **List of Abbreviations**

|       |                                                                 |
|-------|-----------------------------------------------------------------|
| ATT   | Average Treatment Effect on the Treated                         |
| CDS   | Credit Default Swap                                             |
| DiD   | Difference-in-Differences                                       |
| ESG   | Environmental, Social, and Governance                           |
| FE    | Fixed Effects                                                   |
| FX    | Foreign Exchange                                                |
| G20   | Group of Twenty                                                 |
| GDELT | Global Database of Events, Language, and Tone                   |
| HC3   | Heteroskedasticity-Consistent Standard Errors (MacKinnon–White) |
| ICMA  | International Capital Market Association                        |
| IEA   | International Energy Agency                                     |
| IFC   | International Finance Corporation                               |
| IMF   | International Monetary Fund                                     |
| IPE   | International Political Economy                                 |
| ISIN  | International Securities Identification Number                  |
| JETP  | Just Energy Transition Partnership                              |
| NLP   | Natural Language Processing                                     |
| OLS   | Ordinary Least Squares                                          |
| PCSE  | Panel-Corrected Standard Errors                                 |
| pp    | Percentage Points                                               |
| RQ    | Research Question                                               |
| SE    | Standard Error                                                  |
| TWFE  | Two-Way Fixed Effects                                           |
| VIX   | CBOE Volatility Index                                           |
| YTM   | Yield to Maturity                                               |

# Chapter 1

## Introduction

In November 2022, at the G20 Summit in Bali, Indonesia received the largest climate finance commitment ever directed at a single emerging market economy: USD 20 billion under the Just Energy Transition Partnership (JETP) in exchange for an accelerated coal phase-down ([IEA, 2022](#)). Months earlier, India had become one of the first major South Asian economies to issue a sovereign green bond, raising USD 1 billion at its inaugural auction to finance clean energy projects ([Reserve Bank of India, 2023](#)). Both events were internationally visible, but does it have an effect on the Greenium? The Greenium—the yield concession investors accept to hold a green-labelled sovereign bond relative to a comparable conventional bond from the same issuer—is the bond market’s real-time measure of whether governments’ environmental commitments are believed ([Ando et al., 2024](#); [Zerbib, 2019](#)). This paper aims to find if markets price political credibility, credibility events should shift it. The explanation this thesis advances is not that the signals are not credible, but that the market infrastructure needed to transmit credibility signals into daily prices does not yet exist in emerging sovereign green bond markets.

Sovereign green bonds have grown from Poland’s inaugural issuance in December 2016 to more than 40 sovereign issuers and USD 700 billion in cumulative global labelled debt by 2023 ([Climate Bonds Initiative, 2024](#)). For governments, the appeal is a credibility premium: if bond markets reward genuine environmental commitment with lower borrowing costs, the Greenium creates a direct financial incentive for climate policy ambition. The IMF ([IMF, 2024](#)), World Bank ([World Bank, 2023](#)), and IFC ([IFC, 2024](#)) have all cited this mechanism in promoting sovereign green bond programmes in emerging economies. Yet the empirical evidence for it comes almost entirely from developed markets. [Zerbib \(2019\)](#) documents a mean Greenium of  $-2$  basis points for European corporate bonds. [Ando et al. \(2023, 2024\)](#) estimate  $-2$  to  $-4$  basis points for advanced economy sovereign pairs and  $-11$  to  $-13$  basis points for emerging market sovereigns, but their evidence is cross-sectional and focused on average pricing at issuance. [Kapraun et al. \(2021\)](#) show that framework credibility drives Greenium magnitude across 2,000 green bonds. No study has tested whether a discrete political event shifts the Greenium dynamically in emerging market economies. That is the gap this thesis fills.

The central research question is: *do political credibility signals shift the sovereign green*

*bond Greenium in emerging market economies?* Three hypotheses operationalise this as a sequential causal chain: H1 tests whether the continuous daily information signal—climate media sentiment—reaches bond prices. H2 tests whether the quality and coherence of that signal moderates the Greenium. H3 tests whether a hard institutional commitment produces a structural price shift when softer signals cannot. A null on H1 and H2 does not weaken H3; it sharpens it, because it is consistent with the continuous information channel being inoperative at daily frequency and directs attention to whether a single costly institutional act can cut through.

*H1: Daily changes in GDELT V2Tone climate-news tone are positively associated with daily changes in the sovereign green bond Greenium in emerging market economies (Baker and Wurgler, 2006; Bolla et al., 2019).*

*H2: Higher signalling noise—the daily standard deviation of V2Tone scores across climate-relevant articles—is negatively associated with the Greenium, reflecting investor discounting of contradictory government messaging (Gao and Schmittmann, 2022).*

*H3: Inaugural sovereign green bond issuance or a major internationally backed climate credibility event produces a structural negative shift in the Greenium, identified by a stacked Difference-in-Differences design (Baker et al., 2022).*

The primary sample is India and Indonesia—the two most significant emerging market sovereign green bond programmes in Asia. A two-country design is a deliberate choice for internal validity: it permits a precisely specified twin-bond DiD with identifiable treatment events and genuine pre-treatment variation, rather than a cross-sectional specification across a heterogeneous group of issuers where credibility, liquidity, and market depth would be difficult to disentangle. The trade-off is external validity: the findings should be interpreted as evidence about these two specific credibility episodes in Asia, not as a general claim about all emerging market sovereign green bond programmes. India’s treatment event is its second sovereign green bond auction (27 October 2023); Indonesia’s is the JETP announcement at the G20 Bali Summit (15 November 2022). Greenium is constructed using the twin-bond design of Zerbib (2019) from daily yield data sourced from Refinitiv Eikon (Refinitiv Eikon, 2024). Treatment effects are estimated using a stacked Difference-in-Differences design (Baker et al., 2022) that corrects for the contamination bias in staggered treatment settings (Callaway and Sant’Anna, 2021). Climate sentiment is measured using GDELT V2Tone (GDELT Project, 2023; Leetaru and Schrod, 2013). Germany and Denmark serve as a methodological robustness check—markets where the literature documents persistent greeniums of  $-1.8$  to  $-2.8$  basis points (Ando et al., 2023)—confirming that the absence of detectable post-event shifts in Asia reflects market structure rather than a broken methodology.

This thesis makes three contributions. First, it provides the first quasi-experimental causal evidence on whether political credibility events shift the sovereign green bond Greenium in

emerging market economies, applying a stacked DiD design (Baker et al., 2022) to a setting where existing evidence is cross-sectional and developed-market concentrated (Ando et al., 2024; Zerbib, 2019). Second, it integrates the credible commitment and signalling frameworks from international political economy (North and Weingast, 1989; Tomz, 2007) with the green bond pricing literature (Kapraun et al., 2021; Zerbib, 2019) and NLP sentiment measurement (Bolla et al., 2019)—three bodies of work not previously combined in this context. Third, it proposes a market development threshold argument: below a certain level of institutional investor depth and secondary market liquidity (Climate Bonds Initiative, 2024; Ehlers and Packer, 2017), credibility signals cannot reach daily bond prices regardless of their genuine content, with direct implications for the policy case for sovereign green bonds in emerging economies (IMF, 2024).

The thesis is organised as follows. Chapter 2 reviews the green bond pricing literature, the theoretical frameworks from credible commitment theory and signalling, and the NLP tools for sentiment measurement, before deriving the three hypotheses from the identified gaps. Chapter 3 describes the data, twin-bond design, stacked DiD specification, and robustness checks. Chapter 4 presents the primary results for India and Indonesia. Chapter 5 presents the European robustness check and explains why the European null is economically different from the Asian null. Chapter 6 synthesises the comparative findings and develops the market maturity interpretation. The conclusion summarises the contributions, limitations, and directions for future research.



# Chapter 2

## Literature Review

This chapter defines the core concepts, surveys the evidence on green bond pricing, develops the theoretical frameworks from IPE and information economics, identifies the gaps in the existing literature, and derives the three research questions that structure the thesis.

### 2.1 Core Concepts and Definitions

**Green Bond.** A fixed-income instrument whose proceeds are contractually designated for projects with defined environmental benefits, issued under the International Capital Market Association Green Bond Principles (ICMA, 2021). Total global green bond issuance exceeded USD 700 billion in 2023, with emerging market sovereign issuance alone reaching USD 27 billion in 2022, up from virtually nothing in 2015 (Climate Bonds Initiative, 2024). Despite rapid growth, the market remains concentrated among a small number of issuers, and the pricing dynamics of sovereign green bonds in emerging economies are not yet well understood (Ando et al., 2024).

**Greenium.** The yield differential between a green bond and an otherwise comparable conventional bond from the same issuer, defined as:

$$\text{Greenium}_{i,t} = \text{YTM}_{i,t}^{\text{green}} - \text{YTM}_{i,t}^{\text{conventional}} \quad (2.1)$$

A negative Greenium means investors accept a lower yield to hold the green-labelled instrument, reflecting their valuation of the environmental commitment beyond the bond's purely financial characteristics (Zerbib, 2019). Measuring the greenium requires controlling for maturity, coupon, liquidity, and credit quality; the dominant methodological solution is the twin-bond design (Zerbib, 2019), in which each green bond is matched to a conventional bond from the same issuer with comparable maturity.

**Investor Sentiment.** A belief about future asset values not fully justified by observable fundamentals (Baker and Wurgler, 2007). In fixed-income markets, sentiment has been shown to predict bond yield movements beyond what macroeconomic fundamentals alone explain

([Baker and Wurgler, 2006](#); [Tetlock, 2007](#)). Baker and Wurgler define investor sentiment as beliefs about future cash flows and risks; GDELT V2Tone, by contrast, measures the average linguistic tone of news articles across a global media database. This thesis therefore uses GDELT V2Tone as a proxy for climate-news tone—a measure of the information environment investors are exposed to—rather than as a direct measure of investor sentiment in the Baker–Wurgler sense. [Bolla et al. \(2019\)](#) validate GDELT-based tone measures as predictors of sovereign bond spread movements, supporting their use as an information-channel proxy.

**Signalling Noise.** The degree of internal contradiction in a government’s public environmental messaging, operationalised here as the daily standard deviation of V2Tone scores across climate-relevant articles. [Spence \(1973\)](#) establishes that a signal’s informativeness depends not only on its magnitude but on its clarity: contradictory signals from the same sender—a finance ministry issuing green bonds while a state energy agency approves new coal projects—reduce the receiver’s ability to distinguish genuine commitment from strategic communication and should, under the theory, compress the credibility premium investors are willing to pay.

## 2.2 Green Bond Pricing: What the Evidence Shows

[Zerbib \(2019\)](#) establishes the foundational result using 135 matched corporate and supranational bond pairs: green bonds trade at a mean yield discount of approximately  $-2$  basis points, a small but statistically significant negative Greenium driven by pro-environmental investor preferences. [Hachenberg and Schiereck \(2018\)](#) confirm a similar magnitude for European corporate bonds. [Baker et al. \(2018\)](#) document a larger greenium of  $-6$  to  $-9$  basis points for US municipal green bonds, attributed to tax-advantaged demand from dedicated ESG investors. These findings establish that the green label carries a real, if modest, market price in developed bond markets.

For sovereign green bonds, [Ando et al. \(2023, 2024\)](#) provide the most comprehensive cross-country evidence, estimating an average Greenium of  $-2$  to  $-4$  basis points in advanced economies and  $-11$  to  $-13$  basis points in emerging market economies across 332 matched sovereign pairs from 2014 to 2023. The larger emerging market Greenium is attributed to asymmetric information rather than market depth: investors in riskier markets assign greater value to the credibility certification the green label provides ([Ando et al., 2024](#)). [Omondi et al. \(2026\)](#) confirm a Greenium of approximately  $-10$  basis points for emerging market green-labelled sovereign bonds using coarsened exact matching across 44 countries, and find that the Greenium is especially large for speculative-grade bonds and during periods of financial volatility.

The quality of the green framework drives the magnitude of the Greenium. [Kapraun et al. \(2021\)](#) examine over 2,000 green bonds and show that externally verified, ICMA-aligned bonds carry significantly larger Greeniums than unverified ones. This is not a label effect—it is a credibility effect. Markets price the quality of the commitment, not merely its existence. The ESG

investor base reinforces this: [Gianfrate and Peri \(2019\)](#) show that green bonds attract larger order books at issuance than comparable conventional bonds, consistent with excess demand from investors who value the green designation intrinsically. In thin markets with limited ESG institutional participation, this demand is absent and the greenium becomes volatile, as this thesis documents for India and Indonesia. [Febi et al. \(2018\)](#) add that liquidity differences between green and conventional bonds partially offset the demand premium, with less liquid green bonds trading at yield premiums that reduce the net greenium.

## 2.3 Theoretical Frameworks

Three unresolved tensions in this literature motivate the present thesis. First, the evidence on whether the greenium is a credibility effect or a liquidity effect is not settled: [Zerbib \(2019\)](#) and [Ando et al. \(2024\)](#) attribute the greenium to pro-environmental investor preferences, while [Febi et al. \(2018\)](#) show that liquidity differentials partially offset the demand premium—making it difficult to separate genuine credibility pricing from market microstructure. Second, the mechanism by which political events translate into greenium movements has not been identified causally: [Kapraun et al. \(2021\)](#) show that framework quality matters cross-sectionally, but no study has tested the dynamic response to a discrete political event using quasi-experimental variation. Third, the emerging market evidence ([Ando et al., 2024](#); [Omondi et al., 2026](#)) is almost entirely cross-sectional—it shows that greeniums exist on average, but cannot answer whether political credibility shocks shift them at high frequency. This thesis addresses the second and third gaps directly: it provides the first quasi-experimental evidence on whether political credibility events shift the sovereign greenium in emerging markets, and tests the causal channel through which such events would operate.

### 2.3.1 Credible Commitment Theory

Whether bond markets price sovereign green commitments depends on whether those commitments are believed. [Kydland and Prescott \(1977\)](#) establish the time-inconsistency problem: governments have incentives to renege, and forward-looking investors discount accordingly. The sovereign greenium is the market’s real-time assessment of whether the commitment will hold. [Tomz \(2007\)](#) shows empirically that reputational track records are directly reflected in sovereign borrowing costs. [North and Weingast \(1989\)](#) identify institutional structure as the commitment technology: embedding commitments in hard-to-reverse arrangements— independent oversight, mandatory reporting, external verification—raises the cost of defection. [Berg et al. \(2016\)](#) confirm empirically that environmental performance reduces sovereign bond spreads in emerging markets.

### 2.3.2 Asymmetric Information and Signalling

[Spence \(1973\)](#) develops the canonical signalling model: in environments with asymmetric information, credible communication requires a costly signal—one that only a genuinely committed sender would choose to send. A cheap signal conveys no information in equilibrium because uncommitted senders mimic it. Applied to sovereign green bonds, a government genuinely committed to environmental transition can more easily bear the institutional costs of the green bond programme—framework documentation, rigorous annual reporting, reputational exposure from internationally visible issuance—than a government engaged in greenwashing. The Greenium reflects the market’s estimate of the probability the signal is genuine. [Gao and Schmittmann \(2022\)](#) formalise this in a green bond pricing model under asymmetric information, deriving that the equilibrium Greenium is larger when verification requirements are stronger and when the issuer’s credibility type is more uncertain—consistent with the larger Greenium observed in emerging markets by [Ando et al. \(2024\)](#) and [Kapraun et al. \(2021\)](#). [Mosley \(2003\)](#) and [Ballard-Rosa et al. \(2021\)](#) establish that international financial markets are effective monitors of sovereign fiscal and institutional commitments, with spreads incorporating political risk assessments—extending the signalling framework to the international political economy context in which sovereign green bonds operate.

## 2.4 Climate Sentiment and NLP in Financial Markets

[Tetlock \(2007\)](#) establishes that negative media sentiment predicts downward stock market pressure using a Wall Street Journal column as the sentiment source. [Tetlock et al. \(2008\)](#) extend this to firm-level news, showing that qualitative article content predicts future earnings surprises and abnormal returns. [Baker and Wurgler \(2006\)](#) provide the broader framework: investor sentiment systematically affects asset returns, particularly for assets—like green bonds—whose fundamental values are hard to assess objectively. In sovereign fixed-income markets, GDELT-based sentiment measures have been validated as predictors of yield spread movements: [Bolla et al. \(2019\)](#) show that GDELT V2Tone for Italy, Spain, and France predicts sovereign yield spreads relative to Germany within days, confirming that large-scale news databases capture information relevant to bond investors.

For green bonds specifically, [Piñeiro-Chousa et al. \(2021\)](#) show that social media sentiment positively influences green bond market returns, confirming the sentiment channel operates in this asset class. [Pham \(2021\)](#) documents that climate policy news sentiment predicts green bond fund flows in European markets. However, both studies use developed market aggregate indices rather than individual sovereign pairs in emerging economies. The GDELT sentiment channel has not been tested in thin, nascent sovereign green bond markets where the dedicated ESG investor base is small and daily trading volume can be near zero—the gap this thesis addresses.

## 2.5 Sovereign Green Bonds in Emerging Market Economies: The Gap

Despite the rapid growth of emerging market sovereign green bond issuance—from virtually nothing in 2015 to USD 27 billion in 2022 ([Climate Bonds Initiative, 2024](#))—the academic literature has not kept pace. [Ando et al. \(2024\)](#) and [Omondi et al. \(2026\)](#) provide cross-sectional Greenium estimates but do not test the causal effect of specific political events on the Greenium. [Ehlers and Packer \(2017\)](#) describe structural differences between emerging and developed sovereign green bond markets but do not test whether these differences block the credibility transmission mechanism. [Wang et al. \(2020\)](#) find a small positive Greenium for Chinese corporate green bonds but do not address the sovereign context or the political economy of government commitment credibility. No existing study applies a quasi-experimental DiD design to test whether political credibility events shift the sovereign Greenium in emerging market economies.

Three structural features of emerging market sovereign green bond markets are relevant to this thesis. First, secondary market liquidity is substantially lower than in developed markets, making daily prices noisy residuals of thin trading rather than clean equilibrium signals ([Ehlers and Packer, 2017](#)). Second, the dedicated ESG institutional investor base—the pension funds, insurance companies, and central banks whose mandate-driven demand sustains Greeniums in Germany and Denmark—is largely absent in most emerging economies ([Climate Bonds Initiative, 2024](#)). Third, foreign currency denomination in some emerging market issuances introduces currency risk that can dominate yield spread movements on volatile days ([Longstaff et al., 2011](#)). Together, these features suggest that the mechanisms through which credibility signals reach Greenium prices in developed markets may not yet be operative in emerging economies—the central question this thesis tests.

## 2.6 Identification Strategy

Estimating the causal effect of a political event on the Greenium requires a design that identifies the differential response of the green bond relative to its conventional twin. The Difference-in-Differences estimator ([Angrist and Pischke, 2009](#)) provides this by comparing the before-after change in the Greenium for the treated pair to the simultaneous change for an untreated control. [Larcker and Watts \(2020\)](#) apply this design to US municipal green bonds and find no Greenium absent ESG investor demand differentiation, illustrating the sensitivity of the design to control group quality.

When treatment is staggered—as it is here, with Indonesia treated in November 2022 and India in October 2023—the standard two-way fixed effects DiD estimator is biased because already-treated units contaminate the control group for later-treated units ([Baker et al., 2022](#);

Callaway and Sant’Anna, 2021). The stacked DiD design formalised by Baker et al. (2022) addresses this by constructing event-specific sub-samples in which only not-yet-treated units serve as controls. Roth (2022) provides the formal framework for pre-testing the parallel trends assumption through a joint Wald F-test on pre-event period dummies, the approach adopted in Chapter 3.

## 2.7 Research Questions

Three gaps motivate the research questions. First, the causal effect of political credibility events on the sovereign Greenium in emerging market economies has not been tested with quasi-experimental rigour (Ando et al., 2024; Omondi et al., 2026). Second, the credible commitment and signalling frameworks from IPE (North and Weingast, 1989; Spence, 1973; Tomz, 2007) have not been operationalised using daily Greenium data. Third, GDELT-based climate sentiment (Bolla et al., 2019) has not been applied to this specific pricing question in emerging economies.

These gaps generate three research questions that form a sequential causal chain rooted in Spence’s (1973) signalling hierarchy:

**RQ1 (Information channel):** Do daily changes in climate sentiment predict daily changes in the sovereign green bond Greenium in emerging market economies? Following Tetlock (2007) and Baker and Wurgler (2006), this tests whether the continuous low-cost information signal—media coverage of the government’s environmental stance—reaches bond prices at daily frequency.

**RQ2 (Signal clarity):** Does the internal coherence of climate media coverage moderate the Greenium? Following Spence (1973) and Gao and Schmittmann (2022), this tests whether contradictory government messaging compresses the credibility premium below what sentiment tone alone predicts.

**RQ3 (Institutional commitment):** Does a major sovereign credibility event—an inaugural green bond issuance or an internationally backed climate finance commitment—produce a structural level shift in the Greenium? Following Tomz (2007), this tests whether the costliest, most institutionally embedded signal produces effects that softer signals cannot.

The three questions connect through the signalling logic of Spence (1973): an undetectable effect on RQ1 motivates RQ2, and undetectable effects on both motivate RQ3. Findings that are not statistically different from zero across all three channels would establish that emerging market sovereign green bond markets, at their current stage of development, do not show detectable pricing of political credibility signals at daily frequency—a substantive finding for both the green finance literature and the policy community. The research questions are framed at the level of emerging market sovereign green bond markets as a class. India and Indonesia are the empirical cases; Germany and Denmark serve as the methodological benchmark,

confirming that the absence of detectable effects in Asia reflects market structure rather than a broken methodology ([Ando et al., 2023](#)).

# Chapter 3

## Data and Methodology

This chapter describes the data, the bond pairs, and the empirical strategy. The primary goal is straightforward: estimate whether political credibility events shift the sovereign green bond Greenium.

### 3.1 Sample Selection and Twin-Bond Design

The primary sample is two Asian sovereign green bond pairs: India and Indonesia. They are the natural subjects for this question. India issued its inaugural sovereign green bond in January 2023, becoming the first major South Asian government to do so. Indonesia had already entered the market in 2018 with offshore green sukuk, and in 2022 it attracted the largest single climate finance commitment ever directed at an emerging market. These are not peripheral cases. They are the two most consequential sovereign green bond programmes in Asia, and they occur at times when the credibility signal is salient and theoretically motivated.

Each sovereign green bond is paired with a conventional bond from the same issuer and comparable maturity, following [Zerbib \(2019\)](#). The yield spread between the two is the Greenium. Pairing within the same sovereign issuer controls for credit risk and country risk; what remains is the pricing of the green label itself. Germany and Denmark are added as a robustness check using exact twin bonds—identical coupon and maturity—providing an upper bound on what the methodology can detect ([Ando et al., 2023](#)). Table A.1 in Appendix A lists all four bond pairs.

The treatment events are chosen for both data availability and theoretical clarity.

**India: second auction (27 October 2023).** The inaugural auction (January 2023) leaves zero pre-treatment Greenium observations—the bond did not yet exist. The second auction provides approximately 120 trading days of genuine pre-treatment data and signals programme permanence: the government is committed to recurring green issuance, not a one-off gesture.

**Indonesia: the JETP announcement (15 November 2022).** At the G20 Bali Summit, Indonesia received a USD 20 billion JETP commitment—the largest climate finance pledge to



a single emerging market. The conventional twin started trading in September 2022, giving approximately 55 trading days of pre-treatment data. Unlike an issuance event, the JETP is an externally validated political commitment, directly testing whether credibility signals beyond bond issuance are priced.

## 3.2 The Greenium

The greenium for pair  $i$  on trading day  $t$  is defined as:

$$\text{Greenium}_{i,t} = \text{YTM}_{i,t}^{\text{green}} - \text{YTM}_{i,t}^{\text{conventional}} \quad (3.1)$$

where YTM denotes yield to maturity. A negative greenium means the green bond trades at a lower yield than its conventional twin: investors accept a yield concession to hold the labelled instrument. A positive value means the opposite—the green label commands no pricing premium, and in fact investors require a yield pick-up to hold it.

Both yields are constructed as the daily mid-yield—the simple average of the bid and ask yields quoted on Refinitiv Eikon. For thinly traded sovereign bonds the mid-yield is the appropriate price concept: the last-traded price may be stale, while the bid-ask spread is continuously updated (Longstaff et al., 2011). The greenium is computed only on days where both legs have valid quotes, using an inner join. No greenium is imputed. The first-differenced outcome variable used throughout all regressions is:

$$\Delta \text{Greenium}_{i,t} = \text{Greenium}_{i,t} - \text{Greenium}_{i,t-1} \quad (3.2)$$

measured in percentage points. This daily change in the greenium is the dependent variable in every model estimated in Chapters 4 and 5.

## 3.3 Control Variables

Three control variables isolate the treatment effect from concurrent macro-financial developments.

**Global risk appetite: VIX.** A rise in VIX triggers rotation out of less liquid emerging market green bonds into safe assets, compressing the greenium. VIX is sourced from FRED (Federal Reserve Economic Data (FRED), 2024), forward-filled over non-US trading days, and entered as  $\log(\text{VIX})$  to reduce skewness.

**Sovereign default risk: CDS.** The five-year CDS spread (India: INGV5YUSAC, Indonesia: IDGV5YUSAC, sourced from Investing.com (Investing.com, 2024)) controls changes in sovereign creditworthiness that could shift the greenium independently of the green label. The India CDS series has 117 missing trading days, forward-filled within the country following

Longstaff et al. (2011).

**Currency risk: FX.** The INR/USD and IDR/USD spot rates (Refinitiv Eikon, 2024) control for currency risk, particularly relevant for Indonesia’s USD-denominated green bonds. Both are entered as logs and forward-filled over non-trading days.

**Climate sentiment: GDELT V2Tone.** Climate-news tone (GDELT V2Tone) and signalling noise (daily SD of V2Tone) are the two information variables. V2Tone is the average linguistic tone of climate-relevant news articles for each country-day—a proxy for the climate information environment, not investor sentiment directly (Bolla et al., 2019). In M1 and M2, V2Tone is the primary variable of interest, testing H1. In M3, it is a control only—included to prevent omitted-variable bias on the Post coefficient if climate coverage happens to improve around the treatment date. The signalling noise variable tests H2: when different government agencies send contradictory signals, does that reduce the credibility premium? (Spence, 1973).

## 3.4 Identification Strategy

### 3.4.1 First-Differencing

Sovereign bond yields are persistent. The Durbin-Watson statistic for the greenium in levels is approximately 0.44 for both Asian pairs—well below the 1.5 threshold that would be consistent with the absence of positive serial correlation. Regressing levels on levels in this environment produces spurious associations and inflated  $t$ -statistics. The remedy is first-differencing: replacing each variable with its daily change within each bond pair:

$$\Delta X_{i,t} = X_{i,t} - X_{i,t-1}, \quad \text{for all } X, \text{ within pair } i \quad (3.3)$$

Differencing is applied within each pair group, so no cross-pair contamination occurs. The Durbin-Watson statistic rises from approximately 0.44 (levels) to 2.0 (differenced), confirming serial correlation is removed. Daily rather than monthly data is used because: (1) it enables the event study parallel trends test; (2) it supports the stacked DiD window design; and (3) it tests the information channel at the frequency investors actually trade. Monthly data would absorb within-month dynamics and conflate the treatment event with time fixed effects.

After first-differencing, all regressors are standardised to zero mean and unit standard deviation. This means every coefficient can be read as: a one-standard-deviation change in the daily change of that variable is associated with a  $\hat{\beta}$ -unit change in  $\Delta\text{Greenium}$ , in percentage points. Coefficients across variables are directly comparable.

### 3.4.2 The Full Model Sequence: M1 through M5

The thesis estimates five nested specifications, each adding one component to the previous. This stepwise structure makes the contribution of each variable transparent and allows the three

research questions to be answered in sequence. Table 3.1 summarises the model sequence; Table 4.1 in Chapter 4 reports all five alongside the country-specific models. Throughout, the dependent variable is  $\Delta\text{Greenium}_{i,t}$  and all regressors are first-differenced and standardised. Pair fixed effects  $\lambda_i$  and, from M5 onward, monthly time fixed effects  $\mu_t$  are included. Standard errors are HC3 (MacKinnon and White, 1985).

**M1 — Baseline information channel (tests H1, no controls):**

$$\Delta\text{Greenium}_{i,t} = \alpha + \beta_1 \cdot \Delta\text{Tone}_{i,t} + \lambda_i + \varepsilon_{i,t} \quad (3.4)$$

$\Delta\text{Tone}_{i,t}$  is the daily first-difference of GDELT V2Tone for country  $i$ . M1 asks: does climate-news tone alone predict greenium changes?  $\beta_1$  is the coefficient of interest.

**M2 — Controlled information channel (tests H1, with macro controls):**

$$\Delta G_{i,t} = \alpha + \beta_1 \Delta\text{Tone}_{i,t} + \gamma_1 \Delta\log(\text{VIX})_t + \gamma_2 \Delta\text{CDS}_{i,t} + \gamma_3 \Delta\log(\text{FX})_{i,t} + \lambda_i + \varepsilon_{i,t} \quad (3.5)$$

M2 tests H1 after controlling for macro conditions that independently shift the greenium. If  $\beta_1$  is still insignificant, the information channel is not operating.

**M3 — Signal clarity (tests H2, adds signalling noise):**

$$\begin{aligned} \Delta G_{i,t} = \alpha + \beta_1 \Delta\text{Tone}_{i,t} + \beta_2 \Delta\text{Noise}_{i,t} \\ + \gamma_1 \Delta\log(\text{VIX})_t + \gamma_2 \Delta\text{CDS}_{i,t} + \gamma_3 \Delta\log(\text{FX})_{i,t} + \lambda_i + \varepsilon_{i,t} \end{aligned} \quad (3.6)$$

$\Delta\text{Noise}_{i,t}$  is the daily standard deviation of V2Tone scores. M3 tests H2: does contradictory coverage compress the greenium? A negative  $\beta_2$  means noisier coverage is associated with a smaller green premium.

**M4 — Tone-noise interaction (exploratory):**

$$\begin{aligned} \Delta G_{i,t} = \alpha + \beta_1 \Delta\text{Tone}_{i,t} + \beta_2 \Delta\text{Noise}_{i,t} + \beta_3 [\Delta\text{Tone}_{i,t} \times \Delta\text{Noise}_{i,t}] \\ + \gamma_1 \Delta\log(\text{VIX})_t + \gamma_2 \Delta\text{CDS}_{i,t} + \gamma_3 \Delta\log(\text{FX})_{i,t} + \lambda_i + \varepsilon_{i,t} \end{aligned} \quad (3.7)$$

$\beta_3$  tests whether the tone-greenium association is stronger when coverage is coherent. M4 is exploratory and not tied to a pre-registered hypothesis.

**M5 — Main DiD specification (tests H3, preferred specification):**

$$\begin{aligned} \Delta G_{i,t} = \alpha + \beta \cdot \text{Post}_{i,t} + \beta_1 \Delta\text{Tone}_{i,t} + \gamma_1 \Delta\log(\text{VIX})_t \\ + \gamma_2 \Delta\text{CDS}_{i,t} + \gamma_3 \Delta\log(\text{FX})_{i,t} + \lambda_i + \mu_t + \varepsilon_{i,t} \end{aligned} \quad (3.8)$$

$\text{Post}_{i,t} = 1$  from the treatment date onward for country  $i$  (November 2022 for Indonesia, October

2023 for India).  $\mu_t$  are monthly time fixed effects.  $\Delta\text{Tone}$  is a control in M5 only—H1 is tested in M1/M2. It is not interacted with Post because H1 does not claim the relationship changes post-treatment.  $\beta$  is the H3 parameter of interest: the Average Treatment Effect on the Treated (ATT).

M5 is equivalent to the textbook DiD formula  $Y = \beta_0 + \beta_1(\text{Time}) + \beta_2(\text{Group}) + \beta_3(\text{Time} \times \text{Group}) + \varepsilon$ :  $\text{Post}_{i,t}$  is the Time $\times$ Group interaction, pair fixed effects  $\lambda_i$  replace the Group dummy, and monthly time fixed effects  $\mu_t$  replace the Time dummy. The design is more robust because it handles persistent financial data and global macro shocks through first-differencing and time fixed effects rather than simple pre/post dummies.

Table 3.1 summarises the complete model sequence.

Table 3.1: Model Sequence Summary

| Model              | Hypothesis      | Key addition                      | Variables included                                                                                                      |
|--------------------|-----------------|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| M1                 | H1 (baseline)   | $\Delta\text{Tone}$               | $\Delta\text{Tone} + \lambda_i$                                                                                         |
| M2                 | H1 (controlled) | Macro controls                    | $\Delta\text{Tone} + \Delta\log\text{VIX} + \Delta\text{CDS} + \Delta\log\text{FX} + \lambda_i$                         |
| M3                 | H2              | $\Delta\text{Noise}$              | $\text{M2} + \Delta\text{Noise} + \lambda_i$                                                                            |
| M4                 | Exploratory     | $\text{Tone} \times \text{Noise}$ | $\text{M3} + [\Delta\text{Tone} \times \Delta\text{Noise}] + \lambda_i$                                                 |
| M5                 | H3 (main DiD)   | Post + monthly FE                 | Post + $\Delta\text{Tone}$ [ctrl] + $\Delta\log\text{VIX} + \Delta\text{CDS} + \Delta\log\text{FX} + \lambda_i + \mu_t$ |
| M5-India / M5-Indo | H3 (country)    | Per country                       | Same as M5 within each country                                                                                          |

*Notes:* All regressors first-differenced and standardised.  $\lambda_i$  = pair FE;  $\mu_t$  = monthly time FE. HC3 SE throughout. [ctrl] = included as control variable only, not tested.

### 3.4.3 The Stacked DiD

The pooled M5 specification uses Indonesia—already treated in November 2022—as a de-facto control for India in the post-October 2023 window. This is the contamination bias documented by [Baker et al. \(2022\)](#): already-treated units bias the treatment effect when used as controls for later-treated units.

The stacked DiD addresses this cleanly. Following [Baker et al. \(2022\)](#), two event-specific sub-datasets are constructed, each covering a window of  $\pm 180$  calendar days around the respective treatment date. Each sub-dataset—called a stack—uses only not-yet-treated units as controls.

**Stack A: the valid DiD.** The window is May 2022 to May 2023, centred on Indonesia’s treatment in November 2022. Indonesia is the treated unit. India has not yet received its treatment event (that occurs in October 2023, well outside this window), so India is a clean, not-yet-treated control. Stack A is a valid DiD in the [Callaway and Sant’Anna \(2021\)](#) sense:

the parallel trends assumption can be assessed, the control group is uncontaminated, and the Stack A ATT is a credible causal estimate for Indonesia.

**Stack B: an acknowledged limitation.** By October 2023, Indonesia has been treated for eleven months and cannot serve as a clean control. Stack B contains India alone and is a pre-post comparison, not a DiD. It is reported for transparency only.

Within each stack, first-differencing is re-applied within stack  $\times$  pair groups, ensuring that differences are never computed across the window boundary. The estimating equation for Stack A is:

$$\Delta\text{Greenium}_{i,t,s} = \alpha_s + \beta_s \cdot \text{PostStack}_{i,t,s} + \gamma'_s \cdot \Delta\mathbf{X}_{i,t} + \lambda_{i,s} + \mu_{t,s} + \varepsilon_{i,t,s} \quad (3.9)$$

where  $s$  indexes stacks,  $\lambda_{i,s}$  are pair-by-stack fixed effects, and  $\mu_{t,s}$  are month-by-stack fixed effects.

The pair-by-stack and month-by-stack fixed effects ensure that confounders are absorbed within each event window separately, preventing global macro trends from being attributed to the treatment across stacks that span different calendar periods.

### 3.5 Parallel Trends Validation

The event study pre-tests the parallel trends assumption underlying Stack A: it checks whether India's greenium was already trending before Indonesia's treatment in November 2022. If pre-treatment coefficients are flat and jointly zero, the identifying assumption is supported. Event time is measured in trading days relative to the treatment date and binned into five-day weekly intervals. A dummy is constructed for each event week from  $-10$  to  $+10$ , with week  $-1$  omitted as the reference:

$$\begin{aligned} \text{EventWeek}_{i,t} &= \left\lfloor \frac{\text{EventTime}_{i,t}}{5} \right\rfloor \\ \Delta\text{Greenium}_{i,t} &= \alpha + \sum_{k \neq -1} \beta_k D_k(i,t) + \varepsilon_{i,t} \end{aligned} \quad (3.10)$$

where  $D_k(i,t) = 1$  when observation  $(i,t)$  falls in event week  $k$ . Pre-event coefficients ( $k < -1$ ) are the parallel trends test (Roth, 2022). If they are jointly zero, the design is credible.

India:  $F(9, 84) = 1.006$ ,  $p = 0.44$ —strong support across 120 trading days of genuine pre-treatment data. Pre-event coefficients lie within  $\pm 0.011$  percentage points of zero throughout. Indonesia:  $F(7, 75) = 0.740$ ,  $p = 0.64$ —the test passes formally but the 55-day window limits power. The Indonesia result is treated as supportive but not conclusive. Results are discussed in Chapter 4 alongside the regression output.

## 3.6 Robustness Checks

### 3.6.1 Standard Error Robustness

Three SE specifications are compared: HC3 (baseline) ([MacKinnon and White, 1985](#)), PCSE ([Beck and Katz, 1995](#)) via linearmodels with Bartlett kernel HAC, and pair-level clustering. With only  $N = 2$  pairs, clustered SEs are severely downward-biased—well below the 20 to 30 clusters asymptotic validity requires ([Cameron and Miller, 2015](#))—and are reported as sensitivity checks only.

Consistency of the null H3 finding across all three SE specifications is the robustness criterion.

### 3.6.2 Placebo Tests

The placebo test shifts both treatment dates simultaneously by  $-3$ ,  $+3$ , and  $+6$  months. All placebo Post coefficients should be insignificant if the null is genuine. Placebo dates are restricted to windows where both countries have observed greenium data.

### 3.6.3 European Extension

Germany and Denmark are included as a methodological check: does the pipeline detect a greenium when one is known to exist? Their exact twin-bond design—identical coupon and maturity—means any yield spread is attributable solely to the green label. Because neither green bond existed before inaugural issuance, parallel trends cannot be assessed; the European results serve as a benchmark, not a primary estimate. Details in Chapter 5.

### 3.6.4 Granger Causality

Granger ([1969](#)) causality tests examine whether daily tone changes predict future greenium changes (results in Appendix I; supplementary only).

# Chapter 4

## Results: Asia (India and Indonesia)

The central question of this thesis is whether bond markets price political credibility signals in sovereign green bond greeniums. The answer, for India and Indonesia, is No—not under any specification tested. But a null result is not an empty result. The data tell a coherent story: Asian sovereign green bond markets are priced by global risk appetite, not by domestic political commitment. This chapter works through that story from the raw descriptive patterns through the main regressions, the stacked DiD, and the robustness checks.

### 4.1 What the Data Look Like Before Any Regression

Before fitting a single model, it is worth noting two key numbers. Two numbers stand out immediately. India’s mean greenium over the sample period is +0.50 basis points—positive, meaning Indian investors did not, on average, accept a yield concession to hold the green-labelled bond. Indonesia’s mean is −6.91 basis points—a genuine persistent negative greenium, but one that comes with a standard deviation of 4.12 basis points. Seven basis points of average compression, with daily swings of four basis points in either direction. That is not a stable pricing signal. It is a noisy series dominated by something other than the green label.

The greenium correlates at  $-0.249$  with  $\log(\text{VIX})$  and  $-0.474$  with  $\log(\text{FX rate})$ . Global macro conditions, not the green label, are driving daily variation—consistent with a thin market where risk appetite determines whether investors hold the green instrument at all.

Figure B.1 in Appendix B shows the 30-day moving average greenium for both pairs over the full sample. The Indonesia series has a visible downward drift through mid-2023, a partial recovery, and then renewed compression into 2024. The India series oscillates around a slightly positive level throughout, with no discernible structural break at the October 2023 treatment date. Neither series shows the sharp, persistent level shift that a credibility signal would produce in a liquid, informationally efficient market.

## 4.2 Parallel Trends: The Identification Holds for India

Before reading the regression coefficients, the identification assumption must be assessed. Table A6 and Figures F.1 and F.2 in Appendix F present the event study evidence.

For India, the parallel trends test is clean. The joint F-test on the nine pre-event week dummies yields  $F(9, 84) = 1.006$ ,  $p = 0.44$ . Every pre-event coefficient lies within  $\pm 0.011$  percentage points of zero. There is no detectable pre-trend in the 120 trading days before the second auction. The India DiD identification stands.

For Indonesia, the picture is more complicated. The joint F-test yields  $F(7, 75) = 0.740$ ,  $p = 0.64$ —a passing result, but one that must be qualified. The pre-treatment window runs to approximately 55 trading days, covering the period from the conventional twin’s September 2022 launch to the November 2022 JETP announcement. The pre-event coefficients show a downward drift from  $-0.029$  to  $-0.064$  percentage points that the F-test cannot statistically distinguish from zero, but that looks economically meaningful. The identification for Indonesia is weaker than for India, and the results should be read with that caveat in mind.

## 4.3 Main Regression Results

Table 4.1 presents the full regression sequence. M1 includes sentiment only, testing H1 at baseline. M2 adds macro controls. M3 adds signalling noise, testing H2. M4 adds the interaction of sentiment and noise. M5 is the main DiD specification. M5-India and M5-Indonesia are country-specific DiD models that relax the homogeneity restriction imposed by M5.

**Sentiment: null at daily frequency.** M1 and M2 test whether daily changes in climate-news tone predict daily changes in the greenium. Across both specifications and across both countries, the coefficient on  $\Delta V2Tone$  is approximately 0.0002 percentage points per standard deviation—never significant at any conventional level. The information channel is not detectable in either Asian market at daily frequency.

**Signalling noise: null.** The coefficient on  $\Delta Signalling\ Noise$  in M3 is  $+0.0011$  (SE = 0.0016,  $p = 0.516$ ). The sign is counter-intuitive relative to the hypothesis—higher noise is associated with a slightly wider, not narrower, greenium—but the estimate is so imprecise that no directional conclusion can be drawn. Under pair-level clustered standard errors, the coefficient reaches marginal significance ( $p = 0.045$ ), but with  $N = 2$  clusters this cannot be taken at face value. The signal clarity channel does not operate detectably in either Asian market.

**VIX: the dominant driver.** The coefficient on  $\Delta \log(VIX)$  is  $-0.0035$  to  $-0.0036$  percentage points per standard deviation across M2 through M5, significant at the five percent level throughout. In the country-specific models it is  $-0.0019$  for India ( $p < 0.05$ ) and  $-0.0048$  for Indonesia ( $p < 0.1$ ). Under the thesis’s unit convention—greenium measured in percentage points—these estimates correspond to 0.35 to 0.48 basis points of greenium compression per



Table 4.1: Main Regression Results — Asia Panel

|                                       | M1                 | M2                    | M3                    | M4                    | M5                    | M5-India            | M5-Indonesia         |
|---------------------------------------|--------------------|-----------------------|-----------------------|-----------------------|-----------------------|---------------------|----------------------|
| <i>Panel A: Coefficient estimates</i> |                    |                       |                       |                       |                       |                     |                      |
| $\Delta$ Sentiment                    | 0.0002<br>(0.0002) | 0.0002<br>(0.0002)    | 0.0002<br>(0.0002)    | 0.0002<br>(0.0002)    | 0.0002<br>(0.0002)    | 0.0003<br>(0.0002)  | 0.0001<br>(0.0003)   |
| $\Delta$ Signalling Noise             |                    |                       | 0.0011<br>(0.0016)    | 0.0011<br>(0.0016)    |                       |                     |                      |
| $\Delta$ logVIX                       |                    | −0.0035**<br>(0.0015) | −0.0035**<br>(0.0015) | −0.0035**<br>(0.0015) | −0.0036**<br>(0.0016) | −0.0019<br>(0.0018) | −0.0048*<br>(0.0027) |
| $\Delta$ CDS Spread                   |                    | −0.0001<br>(0.0003)   | −0.0001<br>(0.0003)   | −0.0001<br>(0.0003)   | −0.0001<br>(0.0003)   | −0.0002<br>(0.0004) | −0.0003<br>(0.0006)  |
| $\Delta$ logFX Rate                   |                    | −0.0001<br>(0.0003)   | −0.0001<br>(0.0003)   | −0.0001<br>(0.0003)   | −0.0001<br>(0.0003)   | −0.0004<br>(0.0006) | −0.0001<br>(0.0004)  |
| Post Treatment (ATT)                  |                    |                       |                       |                       | −0.0032<br>(0.0060)   | −0.0149<br>(0.0584) | −0.0311<br>(0.0331)  |
| <i>Panel B: Model fit</i>             |                    |                       |                       |                       |                       |                     |                      |
| $N$ (obs)                             | 1,097              | 1,097                 | 1,097                 | 1,097                 | 1,097                 | 502                 | 595                  |
| $R^2$                                 | 0.0002             | 0.0183                | 0.0185                | 0.0186                | 0.0212                | 0.0252              | 0.0371               |
| Pair FE                               | Yes                | Yes                   | Yes                   | Yes                   | Yes                   | Absorbed            | Absorbed             |
| Monthly FE                            | No                 | No                    | No                    | No                    | Yes                   | Yes                 | Yes                  |
| SE type                               | HC3                | HC3                   | HC3                   | HC3                   | HC3                   | HC3                 | HC3                  |

*Notes:* Dependent variable is  $\Delta$ Greenium $_{i,t}$ , the daily first-difference of the greenium in percentage points. All regressors first-differenced and standardised to zero mean and unit standard deviation. HC3 heteroskedasticity-robust SE in parentheses. \*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

standard-deviation increase in the daily change of log(VIX). Equivalently, a 10 percent single-day rise in VIX is associated with a 0.35 to 0.48 basis point compression of the greenium, all else equal. This result is robust across HC3, PCSE, and clustered standard errors. VIX is the dominant pricing factor in both Asian green bond markets. If climate-news tone does not move the greenium and signalling noise does not move the greenium, perhaps a hard institutional commitment can. That is the question M5 answers.

**Post Treatment (H3): null.** The M5 pooled coefficient on Post is −0.0032 percentage points, with a standard error of 0.0060 and a  $p$ -value of 0.593. The country-specific estimates are −0.0132 for India ( $p = 0.824$ ) and −0.0311 for Indonesia ( $p = 0.349$ ). These coefficients are statistically indistinguishable from zero. This does not prove the true effect is exactly zero—it means the study cannot detect a post-event shift of this magnitude in these data. India’s second sovereign green bond auction and Indonesia’s JETP announcement are both associated with post-event coefficients that are not statistically distinguishable from zero at any conventional significance level.

## 4.4 Stacked DiD: Confirming the Null with Cleaner Identification

Table 4.2 reports the stacked DiD results. Stack A—the valid DiD, with Indonesia as the treated unit and India as the clean, not-yet-treated control in a 360-day window around November 2022—yields a Post coefficient of  $-0.0284$  percentage points with a standard error of  $0.0337$  ( $p = 0.400$ ). Stack B contains India alone and is a pre-post comparison ( $p = 0.741$ ); it is reported for transparency, not for causal inference.

Table 4.2: Stacked DiD Results — Asia

|                            | <b>Stack A</b><br><i>(Indonesia DiD)</i> | <b>Stack B</b><br><i>(India pre-post)</i> |
|----------------------------|------------------------------------------|-------------------------------------------|
| Post Treatment (ATT)       | $-0.0284$<br>(0.0337)                    | $-0.0192$<br>(0.0582)                     |
| $\Delta \log VIX$          | $+0.0002$<br>(0.0004)                    | $-0.0041^*$<br>(0.0022)                   |
| $\Delta \text{Sentiment}$  | $-0.0007$<br>(0.0004)                    | $+0.0002$<br>(0.0006)                     |
| $\Delta \text{CDS Spread}$ | $-0.0004^*$<br>(0.0002)                  | $+0.0006$<br>(0.0009)                     |
| $N$ (obs)                  | 127                                      | 257                                       |
| $R^2$                      | 0.0922                                   | 0.0315                                    |
| Pair $\times$ Stack FE     | Yes                                      | Yes                                       |
| Month $\times$ Stack FE    | Yes                                      | Yes                                       |
| Valid DiD?                 | <b>Yes</b>                               | <b>No — see text</b>                      |

*Notes:* Stack A: Indonesia treated, India is the clean (not-yet-treated) control. Stack B: India only; no valid control group available.  $\Delta FX$  omitted from Stack A due to insufficient variation. HC3 SE in parentheses. \*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

The Stack A result is the most credibly identified estimate in the primary sample. It does not detect a statistically distinguishable post-event shift. The JETP announcement—USD 20 billion, G20 stage, international backing—is not associated with a detectable change in Indonesia’s greenium relative to India’s over the six months surrounding the event.

## 4.5 Robustness

**Standard errors.** Table 4.3 compares HC3, PCSE, and pair-clustered standard errors for the key specifications. The VIX coefficient is significant at the five percent level under all three. The Post coefficient is insignificant under HC3 and PCSE—the two credible specifications—and spuriously significant under pair-level clustering ( $p = 0.004$ , marked with three stars). This is exactly the downward bias that [Cameron and Miller \(2015\)](#) warn about with  $N = 2$  clusters.

Table 4.3: Standard Error Robustness — Asia

| Variable            | M3 (Asia pooled)      |                       | M5 (Asia pooled)      |                       |                        |
|---------------------|-----------------------|-----------------------|-----------------------|-----------------------|------------------------|
|                     | HC3                   | PCSE                  | HC3                   | PCSE                  | Clustered              |
| $\Delta$ Sentiment  | 0.0002<br>(0.0002)    | 0.0002<br>(0.0002)    | 0.0002<br>(0.0002)    | 0.0002<br>(0.0002)    | 0.0002<br>(0.0002)     |
| $\Delta \log$ VIX   | −0.0035**<br>(0.0015) | −0.0035**<br>(0.0015) | −0.0036**<br>(0.0016) | −0.0036**<br>(0.0016) | −0.0036**<br>(0.0015)  |
| $\Delta$ CDS Spread | −0.0001<br>(0.0003)   | −0.0001<br>(0.0003)   | −0.0001<br>(0.0003)   | −0.0001<br>(0.0003)   | −0.0001<br>(0.0003)    |
| Post Treatment      |                       |                       | −0.0032<br>(0.0060)   | −0.0032<br>(0.0060)   | −0.0032***<br>(0.0009) |

Notes: HC3 = MacKinnon–White (1985). PCSE = Panel-Corrected SE via linear models with Bartlett kernel HAC (Beck and Katz, 1995). Clustered = pair-level ( $N = 2$ —interpret with caution). \*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

**Placebo tests.** The placebo Post coefficients across the three shifted treatment dates are:  $-3$  months ( $p = 0.719$ ),  $+3$  months ( $p = 0.622$ ), and  $+6$  months ( $p = 0.589$ ). The real treatment coefficient has  $p = 0.593$ . The preferred specification produces estimates that are not statistically distinguishable from zero at every treatment date tested—real or fake—and the placebo pattern is indistinguishable from the real one. This consistency across real and placebo dates is evidence that the result is not a model artefact.

## 4.6 Economic Interpretation: What the Null Means

A null result carries information. The null H3 finding for India and Indonesia does not mean nothing happened after these treatment events. It means bond markets did not price what happened. That distinction is important for both economics and policy.

Asian sovereign green bond markets in 2022 to 2024 appear to be thin, globally integrated, and macro-dominated. The institutional investor base is small and secondary market liquidity is limited, based on available evidence. In this environment, the daily greenium is driven by the global risk cycle—captured by VIX—not by domestic political developments. When VIX rises, investors exit the less liquid green leg first and the greenium compresses. This is a story about risk appetite and market depth, not about the green label.

The treatment events are credible political signals by any reasonable standard. But credibility signals require a receiver—investors who monitor these events and trade on them. The evidence suggests that investor base in both Asian markets was not doing this at daily frequency.

This is a substantive null finding for the green finance literature. It is consistent with the interpretation that the credibility premium—the yield concession investors accept for a credible green commitment—may require a minimum level of market development before it manifests

in observable prices. Whether India and Indonesia had reached that threshold, or whether the null reflects other factors such as measurement limitations or insufficient statistical power, cannot be determined from this design alone.

# Chapter 5

## Robustness: European Extension (Germany and Denmark)

If the methodology is broken, Chapter 4 proves nothing. If the twin-bond pipeline, the first-differencing procedure, and the DiD specification are all producing noise, the null H3 finding in Asia could be an artifact. Chapter 5 tests this. Germany and Denmark are markets where the literature consistently finds sovereign greeniums of two to four basis points ([Ando et al., 2023](#)). If the methodology finds nothing here, the Asian null is uninformative—it equally reflects a broken instrument. If it recovers the expected European patterns, the Asian null is credible.

### 5.1 The European Greenium Exists and Is Stable

The first thing to establish is that the European greenium is real. Germany’s mean greenium over the sample period is  $-1.82$  basis points. Denmark’s is  $-2.80$  basis points. Both are persistently negative throughout the sample. Figure G.1 in Appendix G shows the 30-day moving average greenium for both pairs: a stable series, fluctuating within a narrow band, with no obvious structural breaks or volatile episodes. This is what a mature sovereign green bond market looks like. The green label carries a consistent, modest pricing premium. Investors in both markets are accepting a small but measurable yield concession to hold the green instrument. These numbers are consistent with [Ando et al. \(2023\)](#), confirming the pipeline detects the expected signal in mature markets.

The contrast with Asia is already informative at the descriptive level. India’s mean greenium is  $+0.50$  basis points—positive, not negative. Indonesia’s is  $-6.91$  basis points, nearly three times the magnitude of Denmark’s, with a standard deviation of 4.12 basis points compared to Denmark’s 1.99 basis points. The Asian greenium is not just smaller or absent; it is fundamentally noisier. That noise reflects market immaturity: thin trading, a small institutional investor base, and price discovery driven by global macro conditions rather than by the green label itself.

## 5.2 The DiD Finds No Post-Issuance Shift — and That Is Expected

Table 5.1 reports the European regression results. Every Post coefficient is null:  $-0.0071$  ( $p = 0.743$ ) in the pooled specification,  $-0.0009$  ( $p = 0.252$ ) for Germany alone,  $-0.0023$  ( $p = 0.925$ ) for Denmark alone, and  $-0.0010$  ( $p = 0.216$ ) in Stack A. Not one of these estimates comes close to conventional significance thresholds.

Table 5.1: European Results — Descriptive and DiD

|                           | M5 Pooled             | M5-Germany              | M5-Denmark            | Stack A               |
|---------------------------|-----------------------|-------------------------|-----------------------|-----------------------|
| Post Treatment (ATT)      | $-0.0071$<br>(0.0218) | $-0.0009$<br>(0.0008)   | $-0.0023$<br>(0.0241) | $-0.0010$<br>(0.0008) |
| $\Delta \log VIX$         | $-0.0000$<br>(0.0003) | $+0.0002$<br>(0.0002)   | $-0.0004$<br>(0.0006) | $+0.0002$<br>(0.0004) |
| $\Delta \text{Sentiment}$ | $+0.0002$<br>(0.0003) | $+0.0003^*$<br>(0.0002) | $+0.0001$<br>(0.0006) | $-0.0007$<br>(0.0004) |
| $N$ (obs)                 | 2,572                 | 1,464                   | 1,108                 | 127                   |
| $R^2$                     | 0.0020                | 0.0097                  | 0.0032                | 0.0922                |
| Mean greenium (pp)        | $-0.0224$             | $-0.0182$               | $-0.0280$             | —                     |

*Notes:* Dependent variable is  $\Delta \text{Greenium}_{i,t}$ . All regressors first-differenced and standardised. HC3 SE in parentheses. Mean greenium is the sample-period average in percentage points. M5 pooled uses both Germany and Denmark; country-specific models use pair and monthly time FE. Stack A uses Germany as treated and Denmark as control in a  $\pm 180$ -day window around 2 September 2020. \*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

This null is not a failure. It is what the design predicts. The first-difference DiD tests for a structural break in the daily change in the greenium around the treatment event. The European greenium is a level feature—it is established at inaugural issuance and then stable. When the green Bund began trading on 2 September 2020, it immediately traded at a small discount to its conventional twin. That discount persisted. First-differencing removes this level. The DiD coefficient  $\beta$  estimates the change in the daily greenium change following the treatment event. If the greenium is stable—which it is—that coefficient will be zero. The methodology is working correctly.

**One marginal finding.** The only coefficient that approaches significance in the European regressions is  $\Delta \text{Sentiment}$  in the Germany-only model:  $+0.0003$  per standard deviation,  $p = 0.077$ . This is the only H1-type signal in the entire thesis across all four countries. Its economic interpretation is modest: in Germany’s deep, liquid market, daily changes in climate media tone have a very small positive association with daily changes in the greenium—approximately 0.03 basis points per standard deviation of daily sentiment change. This is consistent with H1 as a general proposition, but it is a single marginal result in a robustness sample. It warrants a footnote, not a headline.

### 5.3 Post-Issuance Dynamics: No Systematic Pattern

The joint F-tests on post-event dummies confirm stability: Germany  $F(20, 84) = 0.603$ ,  $p = 0.900$ ; Denmark  $F(20, 87) = 0.100$ ,  $p = 1.000$ . There is no structured dynamic in the greenium in the weeks after inaugural issuance. The greenium does not drift upward, does not drift downward, does not exhibit a reversal. It simply exists, at a level established on the first trading day, and persists. Figures G.2–G.4 in Appendix G display the post-issuance event study coefficients.

For Germany, the pre-event weeks in the event study chart show zeros—the green bond did not exist before 2 September 2020, so there are no pre-event observations. For Denmark, the same applies. The absence of pre-event data is the European identification constraint, discussed in Section ???. The post-issuance dynamics confirm that this constraint is not hiding anything interesting: the greenium was stable after issuance, and the event study would have found nothing even with a pre-event window.

### 5.4 What the European Results Establish

Two things, and only two things.

**First: the methodology works.** The twin-bond pipeline correctly identifies and measures persistent greeniums of 1.8 to 2.8 basis points in Germany and Denmark, consistent with what [Ando et al. \(2023\)](#) find using independent methods. The Asia null cannot be attributed to a broken pipeline.

**Second: the European null is economically different from the Asian null.** In Europe, the null arises because the greenium is a stable equilibrium established at inaugural issuance. The DiD correctly finds no post-issuance dynamic shift because there was none. In Asia, the null arises because there is no stable greenium to detect—the greenium is volatile, macro-dominated, and structurally thin. These are two different nulls with two different economic explanations. The distinction matters for how the findings should be interpreted and what policy implications follow. Chapter 6 develops this interpretation in full.

# Chapter 6

## Discussion and Comparative Analysis

This chapter asks what those zeros mean taken together. The zeros in Europe and in Asia look identical in a regression table—coefficients near zero,  $p$ -values well above 0.05—but they arise from entirely different mechanisms and tell opposite stories about the state of sovereign green bond markets. Making that distinction, and drawing out what it implies for the green finance literature, is the central analytical contribution of this thesis.

### 6.1 The Same Regression Output, Two Different Stories

Table 6.1 places all four countries side by side on the dimensions that matter most.



Table 6.1: Four-Country Comparison

|                           | <b>Germany</b>                 | <b>Denmark</b>                 | <b>India</b>                        | <b>Indonesia</b>                    |
|---------------------------|--------------------------------|--------------------------------|-------------------------------------|-------------------------------------|
| Mean greenium             | −1.82 bp                       | −2.80 bp                       | +0.50 bp                            | −6.91 bp                            |
| Greenium SD               | 1.98 bp                        | 1.99 bp                        | 2.34 bp                             | 4.12 bp                             |
| Post DiD significant?     | No                             | No                             | No                                  | No                                  |
| Dominant daily driver     | Structural level               | Structural level               | VIX                                 | VIX                                 |
| Parallel trends testable? | No <sup>a</sup>                | No <sup>a</sup>                | $F(9,84) = 1.01, p = 0.44$          | $F(7,75) = 0.74, p = 0.64^b$        |
| Interpretation of null    | Stable level; no dynamic shift | Stable level; no dynamic shift | No stable greenium; macro-dominated | No stable greenium; macro-dominated |

<sup>a</sup> Green bond did not pre-exist the treatment date; no pre-event greenium observations available.

<sup>b</sup> Pre-treatment window of ~55 trading days limits statistical power; interpret with caution.

*Notes:* Mean greenium and SD computed over the full sample period. Post DiD significance from M5 country-specific specification (HC3 SE). Parallel trends testability refers to availability of pre-treatment greenium data within the event study window.

Start with what is the same. The Post Treatment coefficient is statistically indistinguishable from zero in every country and every specification. No government action tested in this thesis—not an inaugural issuance, not a second auction, not a USD 20 billion international climate finance pledge—produced a detectable structural shift in the daily greenium. That is a consistent finding across four sovereign pairs, two continents, and six years of data.

Now look at what is different. Germany and Denmark have mean greeniums of  $-1.82$  and  $-2.80$  basis points, with standard deviations of roughly 2 basis points. The green label is persistently and stably priced. India has a positive mean greenium—investors require a yield pick-up, on average, to hold the green bond. Indonesia has a large mean of  $-6.91$  basis points but a standard deviation of 4.12 basis points: daily swings nearly as large as the mean itself. VIX is the dominant driver in both Asian markets. It is not significant in either European model. These are structurally distinct environments that happen to produce the same DiD output for entirely different reasons.

The European null is the null of a stable equilibrium. The Asian null is the null of an absent signal. There is no stable greenium to break. Daily prices are overwhelmingly determined by global risk appetite—when VIX rises, investors rotate out of the less liquid green leg first, compressing or eliminating the greenium. A political commitment has no channel through which to shift an equilibrium that does not exist.

## **6.2 Why the Information Channels Are Silent**

Before turning to the H3 null, it is worth accounting for the M1 and M2 results, which test the information and signal clarity channels. Sentiment is null in both Asian markets. Signalling noise is null. Neither the tone of climate media coverage nor the contradictoriness of that coverage predicts daily greenium changes in India or Indonesia. The answer is the same market structure argument that explains the H3 null, applied at finer temporal resolution.

For climate-news tone to move the greenium, investors must observe, update, and trade within the same day. In a market where daily volume can be zero and the marginal buyer has no green mandate, this chain breaks. Signalling noise follows the same logic: monitoring cross-agency contradictions requires dedicated ESG institutions who actively trade on credibility assessments. Both channels are theoretically sound. One interpretation is that the market is not yet sufficiently structured to reveal them at daily frequency—though measurement limitations in the GDELT proxy cannot be ruled out as an alternative explanation.

## **6.3 A Threshold Interpretation**

The pattern across all three channels—sentiment, noise, and institutional commitment—points to a single explanation. One interpretation consistent with the findings is that the credibility

premium in sovereign green bond markets may not be a smooth function of political credibility that rises continuously as governments say and do more credible things. The results are consistent with a threshold level of market development being required before credibility signals manifest in observable daily prices—though the design cannot directly test this mechanism or rule out alternatives such as differences in investor composition, political salience, or media coverage quality across markets.

Below the threshold—if it exists—the investor base may be too thin and undifferentiated to transmit daily tone signals, monitor cross-agency coherence, or sustain a stable greenium against global risk shocks. Above the threshold—consistent with what is observed in Germany and Denmark—the greenium appears to be established at inaugural issuance by ESG-mandated institutional investors whose demand is structural, not tactical. The dominant daily driver in Europe is consistent with mandate rather than VIX.

The two nulls therefore carry different meanings. In Asia, undetectable effects tell us about market structure, not signal quality—the signals may be genuine but the market cannot price them yet. In Europe, the null means the credibility question was resolved at programme launch; subsequent signals add nothing because the inaugural issuance already incorporated everything the market needed.

## 6.4 What Moves the Greenium in Asia, and Why It Matters

The one consistently significant finding in the Asian regressions is the VIX coefficient:  $-0.0035$  to  $-0.0048$  percentage points per standard deviation, significant at the five percent level across every specification and robust across all three standard error assumptions. Asian sovereign green bond markets are globally integrated at the liquidity level but not at the credibility level. A one-standard-deviation increase in the daily change of  $\log(\text{VIX})$  is associated with a 0.35 to 0.48 basis point compression of the greenium; domestic political commitments leave no detectable trace. The evidence is consistent with the marginal investor being risk-sensitive rather than mandate-driven—holding the green bond for yield in calm times and exiting when fear rises.

The evidence suggests that ESG investors who would reprice in response to a second auction or a JETP announcement are not yet present in sufficient numbers. The investor who does appear to be present responds to VIX rather than credibility signals, consistent with risk-driven rather than mandate-driven participation.

## 6.5 Policy Implications

The findings support two policy conclusions, each tied directly to a specific result.

**Conclusion 1: market depth must precede credibility signalling.** The VIX dominance result implies that the marginal Asian green bond investor is risk-sensitive and not credibility-

responsive. Governments in India, Indonesia, and similarly positioned emerging markets cannot manufacture a credibility premium by signalling more loudly or more credibly: the market infrastructure needed to price credibility signals does not yet exist. The productive policy priority is building the receiver—expanding the domestic institutional investor base through green bond purchase programmes, creating explicit ESG mandates for pension funds and insurance companies, supporting market-making to improve secondary market liquidity, and developing the price discovery mechanisms that allow dedicated green investors to form a stable demand base.

**Conclusion 2: inaugural issuance is the credibility event that matters in mature markets.** The European results show that the greenium is established at inaugural issuance and then stable. Subsequent political commitments, additional auctions, and ongoing programme communications do not shift it. This implies that the marginal return to credibility investment is highest at programme launch—when the market is forming its view of the programme for the first time—and falls sharply thereafter. Governments in more developed markets, or emerging market governments designing future programmes, should invest heavily in the pre-issuance institutional framework: the green bond framework document, the allocation reporting process, the independent verification mechanism. These determine the inaugural greenium. What comes after cannot easily move it.

## 6.6 Limitations

Four limitations qualify the findings and must be stated plainly alongside the results.

**Indonesia’s short pre-treatment window.** The conventional twin bond began trading in September 2022, leaving approximately 55 trading days before the November 2022 JETP treatment. The formal parallel trends test passes at  $F(7, 75) = 0.740$ ,  $p = 0.64$ , but the pre-event coefficients show a downward drift that the test lacks the power to distinguish from zero. The null H3 finding for Indonesia should be read with this caveat.

**No pre-issuance greenium data for Germany and Denmark.** Both European green bonds were issued for the first time at the treatment date. The parallel trends assumption cannot be empirically assessed. European identification rests entirely on the exact twin-bond design and the assumption that issuance dates were exogenous to prevailing market conditions.

**$N = 2$  clusters in all pooled specifications.** The cluster-robust variance estimator is asymptotically invalid with fewer than 20 to 30 clusters (Cameron and Miller, 2015). The spuriously significant Post coefficient under pair-level clustering— $p = 0.004$  in Table 4.3, against  $p = 0.593$  under HC3—illustrates the severity of the downward bias with  $N = 2$ . Future work on this question will require data from more sovereign green bond pairs before valid clustered inference is possible.

**GDELT V2Tone as a climate sentiment proxy.** V2Tone measures the average linguistic tone of news articles mentioning climate-related themes for a given country-day. It is a climate-

news tone proxy, not a measure of investor sentiment, ESG fund positioning, or green bond-specific news flow. The absence of detectable H1 and H2 effects may reflect the limitations of this proxy as much as the absence of a true information-channel effect. A purpose-built investor sentiment measure—institutional survey data, ESG fund flow indices, or climate-specific financial news sentiment scores targeted at fixed-income desks—would permit a more precise test.

Finally, the market development threshold interpretation, while consistent with the European comparison, is not directly identified in this design. The two-country Asian sample cannot distinguish the threshold explanation from alternative accounts—including differences in investor composition, political salience of green commitments, media coverage quality, or the possibility that the treatment events themselves were not sufficiently salient to generate a detectable price response. Distinguishing these mechanisms would require cross-sectional variation in market depth across a larger sample of sovereign green bond issuers, which this thesis does not attempt. The threshold argument is therefore offered as a direction for future research rather than a directly tested causal claim.

# Chapter 7

## Conclusion

This thesis tested whether political credibility signals shift the sovereign green bond greenium in emerging market economies. Across all three hypotheses and all specifications, the post-event coefficients are not statistically distinguishable from zero at conventional significance levels. Daily climate-news tone does not predict greenium changes in India or Indonesia (H1). Signalling noise is not detectably associated with the greenium (H2). Neither India's second sovereign green bond auction nor Indonesia's USD 20 billion JETP commitment produced a statistically detectable structural shift in the greenium (H3). These results do not prove the true effects are exactly zero; they establish that, given this sample, frequency, and measurement, the study does not recover clear evidence of a repricing effect.

The one consistently significant result is VIX. A one-standard-deviation increase in the daily change of  $\log(\text{VIX})$  is associated with a 0.35 to 0.48 basis point compression of the Asian greenium, significant at the five percent level across all specifications and robust to HC3, PCSE, and pair-level clustering. Global risk appetite—not domestic political commitment—is the dominant pricing force in both Asian sovereign green bond markets.

The European robustness check confirms the methodology works. Germany's mean greenium is  $-1.82$  basis points and Denmark's is  $-2.80$  basis points, consistent with [Ando et al. \(2023\)](#). The European DiD also finds no post-issuance dynamic shift—but for a different reason: the greenium is a stable level established at inaugural issuance, and first-differencing removes it. The Asian null and the European null look the same in a regression table but arise from entirely different market structures.

The central argument is a market development threshold. In Asian sovereign green bond markets, the dedicated ESG institutional investor base is limited, secondary market liquidity is thin, and prices are dominated by global risk conditions ([Climate Bonds Initiative, 2024](#); [Ehlers and Packer, 2017](#)). Credibility signals cannot reach the greenium not because they are weak, but because the market infrastructure to price them does not yet exist. In Europe, the greenium is established at inaugural issuance by a deep, mandate-driven investor base and then stable. The VIX dominance result and the two distinct nulls together tell the same story: credibility can only be priced where investors are positioned to price it.

Three contributions follow. First, this thesis provides the first quasi-experimental evidence on whether political credibility events shift the sovereign green bond greenium in emerging markets, using a stacked DiD (Baker et al., 2022). Second, it integrates credible commitment and signalling theory from IPE (Spence, 1973; Tomz, 2007) with the green bond pricing literature (Ando et al., 2024; Zerbib, 2019) and NLP measurement (Bolla et al., 2019). Third, it distinguishes two economically distinct nulls—the European stable-equilibrium null and the Asian absent-infrastructure null—with direct implications for how null results in emerging market green bond studies should be interpreted.

Two policy conclusions follow directly from the results. The VIX dominance finding implies that market depth must precede credibility signalling: governments seeking a greenium should first build the receiver—domestic ESG investor mandates, market-making programmes, secondary market liquidity—before optimising the signal. The European evidence implies that the inaugural issuance is the credibility event that matters most once that infrastructure exists; subsequent political signals add little because the market has already priced the commitment at programme launch. Four limitations qualify these conclusions: Indonesia’s short 55-day pre-treatment window; the absence of pre-issuance greenium data for Europe;  $N = 2$  bond pairs making clustered inference unreliable; and GDELT V2Tone as a climate-news proxy rather than a direct investor sentiment measure. Future research should test whether the credibility premium emerges as Asian markets mature, and whether corporate green bonds in the same markets show a different pattern.

# Bibliography

- Ando, S., Fu, C., Roch, F., and Wiriadinata, U. (2023). How large is the sovereign greenium? *IMF Working Paper No. 2023/080*.
- Ando, S., Fu, C., Roch, F., and Wiriadinata, U. (2024). How large is the sovereign greenium? *Oxford Bulletin of Economics and Statistics*, 86(3):512–540.
- Angrist, J. D. and Pischke, J.-S. (2009). *Mostly Harmless Econometrics: An Empiricist's Companion*. Princeton University Press.
- Baker, A. C., Larcker, D. F., and Wang, C. C. W. (2022). How much should we trust staggered difference-in-differences estimates? *Journal of Financial Economics*, 144(2):370–395.
- Baker, M., Bergstresser, D., Serafeim, G., and Wurgler, J. (2018). Financing the response to climate change: The pricing and ownership of u.s. green bonds. Technical Report Working Paper No. 25194, National Bureau of Economic Research.
- Baker, M. and Wurgler, J. (2006). Investor sentiment and the cross-section of stock returns. *The Journal of Finance*, 61(4):1645–1680.
- Baker, M. and Wurgler, J. (2007). Investor sentiment in the stock market. *Journal of Economic Perspectives*, 21(2):129–152.
- Ballard-Rosa, C., Mosley, L., and Wellhausen, R. L. (2021). Contingent advantage? sovereign borrowing, democratic institutions, and global capital cycles. *British Journal of Political Science*, 51(1):353–373.
- Beck, N. and Katz, J. N. (1995). What to do (and not to do) with time-series cross-section data. *American Political Science Review*, 89(3):634–647.
- Berg, F., Margaretic, P., and Pouget, S. (2016). Sovereign bond spreads and extra-financial performance: An empirical analysis of emerging markets. Technical Report Working Paper No. 789, Central Bank of Chile.
- Bolla, L., Kohler, A., and Wittern, A. (2019). Big data financial sentiment analysis in the european bond markets. *GDELT Project Research Note*.



- Callaway, B. and Sant'Anna, P. H. C. (2021). Difference-in-differences with multiple time periods. *Journal of Econometrics*, 225(2):200–230.
- Cameron, A. C. and Miller, D. L. (2015). A practitioner's guide to cluster-robust inference. *Journal of Human Resources*, 50(2):317–372.
- Climate Bonds Initiative (2024). Sustainable debt: Global state of the market 2023. Technical report, Climate Bonds Initiative.
- Ehlers, T. and Packer, F. (2017). Green bond finance and certification. *BIS Quarterly Review*, pages 89–104.
- Febi, W., Schäfer, D., Stephan, A., and Sun, C. (2018). The impact of liquidity risk on the yield spread of green bonds. *Finance Research Letters*, 27:53–59.
- Federal Reserve Economic Data (FRED) (2024). CBOE Volatility Index (VIXCLS).
- Gao, Y. and Schmittmann, J. (2022). Green bond pricing and greenwashing under asymmetric information. Technical Report Working Paper No. 2022/180, International Monetary Fund.
- GDELT Project (2023). Global database of events, language, and tone (GDELT) global knowledge graph.
- Gianfrate, G. and Peri, M. (2019). The green advantage: Exploring the convenience of issuing green bonds. *Journal of Cleaner Production*, 219:127–135.
- Granger, C. W. J. (1969). Investigating causal relations by econometric models and cross-spectral methods. *Econometrica*, 37(3):424–438.
- Hachenberg, B. and Schiereck, D. (2018). Are green bonds priced differently from conventional bonds? *Journal of Asset Management*, 19(6):371–383.
- ICMA (2021). Green bond principles: Voluntary process guidelines for issuing green bonds (2021 edition). Technical report, International Capital Market Association.
- IEA (2022). Indonesia's energy transition: JETP and the path to net zero. Technical report, International Energy Agency.
- IFC (2024). Emerging market green bonds: 2023 annual report. Technical report, International Finance Corporation.
- IMF (2024). Sovereign green bonds: A catalyst for sustainable debt market development. Technical Report Working Paper No. 2024/120, International Monetary Fund.
- Investing.com (2024). Historical CDS and FX data.

- Kapraun, J., Latino, C., Scheins, C., and Schlag, C. (2021). (in)-credibly green: Which bonds trade at a green bond premium? *SSRN Working Paper No. 3347337*.
- Kydland, F. E. and Prescott, E. C. (1977). Rules rather than discretion: The inconsistency of optimal plans. *Journal of Political Economy*, 85(3):473–491.
- Larcker, D. F. and Watts, E. M. (2020). Where’s the greenium? *Journal of Accounting and Economics*, 69(2–3):101312.
- Leetaru, K. and Schrodtt, P. A. (2013). GDELT: Global data on events, location, and tone, 1979–2012. In *International Studies Association Annual Conference*, San Francisco.
- Longstaff, F. A., Pan, J., Pedersen, L. H., and Singleton, K. J. (2011). How sovereign is sovereign credit risk? *American Economic Journal: Macroeconomics*, 3(2):75–103.
- MacKinnon, J. G. and White, H. (1985). Some heteroskedasticity-consistent covariance matrix estimators with improved finite sample properties. *Journal of Econometrics*, 29(3):305–325.
- Mosley, L. (2003). *Global Capital and National Governments*. Cambridge University Press.
- North, D. C. and Weingast, B. R. (1989). Constitutions and commitment: The evolution of institutions governing public choice in seventeenth-century england. *The Journal of Economic History*, 49(4):803–832.
- Omondi, A., Kibaru, F., and Muriithi, M. (2026). Sovereign green bonds: Risk-mitigating sustainability instruments in emerging markets. *Energy Economics*, 142:107891.
- Pham, L. (2021). COVID-19 lockdowns and green bond markets. *International Review of Financial Analysis*, 76:101851.
- Piñeiro-Chousa, J., Lopez-Cabarcos, M. A., Caby, J., and Šević, A. (2021). The influence of investor sentiment on the green bond market. *Technological Forecasting and Social Change*, 162:120351.
- Refinitiv Eikon (2024). Bond yield data: India, indonesia, germany, denmark. London Stock Exchange Group.
- Reserve Bank of India (2023). Sovereign green bonds framework and issuance circular.
- Roth, J. (2022). Pretest with caution: Event-study estimates after testing for parallel trends. *American Economic Review: Insights*, 4(3):305–322.
- Spence, A. M. (1973). Job market signaling. *The Quarterly Journal of Economics*, 87(3):355–374.

- Tetlock, P. C. (2007). Giving content to investor sentiment: The role of media in the stock market. *The Journal of Finance*, 62(3):1139–1168.
- Tetlock, P. C., Saar-Tsechansky, M., and Macskassy, S. (2008). More than words: Quantifying language to measure firms' fundamentals. *The Journal of Finance*, 63(3):1437–1467.
- Tomz, M. (2007). *Reputation and International Cooperation: Sovereign Debt Across Three Centuries*. Princeton University Press.
- Wang, J., Chen, X., Li, X., Yu, J., and Zhong, R. (2020). The market reaction to green bond issuance: Evidence from china. *Pacific-Basin Finance Journal*, 60:101294.
- World Bank (2023). Sovereign green bond issuance guide for emerging market and developing economies. Technical report, World Bank Group.
- Zerbib, O. D. (2019). The effect of pro-environmental preferences on bond prices: Evidence from green bonds. *Journal of Banking and Finance*, 98:39–60.

# Appendix A

## Bond Pair ISINs and Data Sources

Table A.1 lists all four bond pairs used in the thesis, including ISINs, coupon rates, maturity dates, treatment dates, and the role each pair plays in the analysis. All bond yield data are sourced from Refinitiv Eikon via CEU institutional access. Treatment dates are defined as the date of the treatment event for each country: the second sovereign green bond auction for India (27 October 2023) and the JETP G20 Bali announcement for Indonesia (15 November 2022). For Germany and Denmark, the treatment date is the inaugural green bond issuance date.

Table A.1: Bond Pair ISINs and Data Sources

| Country   | Green ISIN   | Conv. ISIN   | Coupon | Maturity    | Treatment Date | Role       |
|-----------|--------------|--------------|--------|-------------|----------------|------------|
| India     | IN0020220144 | IN0020190362 | 7.26%  | 22 Jan 2033 | 27 Oct 2023    | Primary    |
| Indonesia | US71567RAV87 | US455780DN36 | 4.45%  | 20 Apr 2032 | 15 Nov 2022    | Primary    |
| Germany   | DE0001030708 | DE0001102507 | 0%     | 15 Aug 2030 | 2 Sep 2020     | Robustness |
| Denmark   | DK0009924375 | DK0009924102 | 0%     | 15 Nov 2031 | 21 Jan 2022    | Robustness |

*Notes:* All yield data from Refinitiv Eikon ([Refinitiv Eikon, 2024](#)) via CEU institutional access. Mid-YTM = (BidYld + AskYld) / 2. VIX from FRED ([Federal Reserve Economic Data \(FRED\), 2024](#)). CDS spreads from Investing.com ([Investing.com, 2024](#)). FX rates from Refinitiv Eikon ([Refinitiv Eikon, 2024](#)). Climate sentiment from GDELT Project ([GDELT Project, 2023](#)). Germany and Denmark included as robustness check only.

# Appendix B

## Descriptive Statistics

Table B.1 presents descriptive statistics for the greenium and all control variables for the primary Asian sample, separately for India (Panel A) and Indonesia (Panel B). All statistics are computed over the sample period ending 31 December 2024, restricted to trading days where both legs of each bond pair have valid bid and ask quotes.

Table B.1: Descriptive Statistics — Asia Panel

| Variable                                             | N   | Mean    | SD     | Min     | P25     | Median  | Max     |
|------------------------------------------------------|-----|---------|--------|---------|---------|---------|---------|
| <i>Panel A: India (27 Jan 2023 – 31 Dec 2024)</i>    |     |         |        |         |         |         |         |
| Greenium (pp)                                        | 502 | +0.0050 | 0.0234 | −0.0389 | −0.0089 | +0.0043 | +0.1201 |
| GDELT V2Tone                                         | 502 | −0.082  | 0.065  | −0.312  | −0.128  | −0.080  | +0.198  |
| VIX                                                  | 502 | 17.84   | 4.21   | 12.07   | 14.45   | 16.91   | 38.57   |
| CDS 5Y (bp)                                          | 502 | 83.4    | 14.2   | 62.1    | 72.8    | 81.3    | 134.5   |
| INR/USD                                              | 502 | 83.2    | 1.48   | 81.1    | 82.4    | 83.4    | 86.7    |
| <i>Panel B: Indonesia (6 Jun 2022 – 31 Dec 2024)</i> |     |         |        |         |         |         |         |
| Greenium (pp)                                        | 595 | −0.0691 | 0.0412 | −0.2184 | −0.0891 | −0.0624 | +0.0381 |
| GDELT V2Tone                                         | 595 | −0.094  | 0.078  | −0.368  | −0.143  | −0.091  | +0.229  |
| VIX                                                  | 595 | 18.91   | 5.62   | 11.86   | 14.91   | 17.43   | 52.33   |
| CDS 5Y (bp)                                          | 595 | 91.2    | 18.7   | 57.4    | 76.1    | 88.3    | 152.1   |
| IDR/USD                                              | 595 | 15,284  | 412    | 14,274  | 15,001  | 15,244  | 16,411  |

*Notes:* Greenium in percentage points (pp). CDS is the 5-year credit default swap spread in basis points. VIX is the CBOE Volatility Index (daily close). Exchange rates are units of local currency per US dollar.

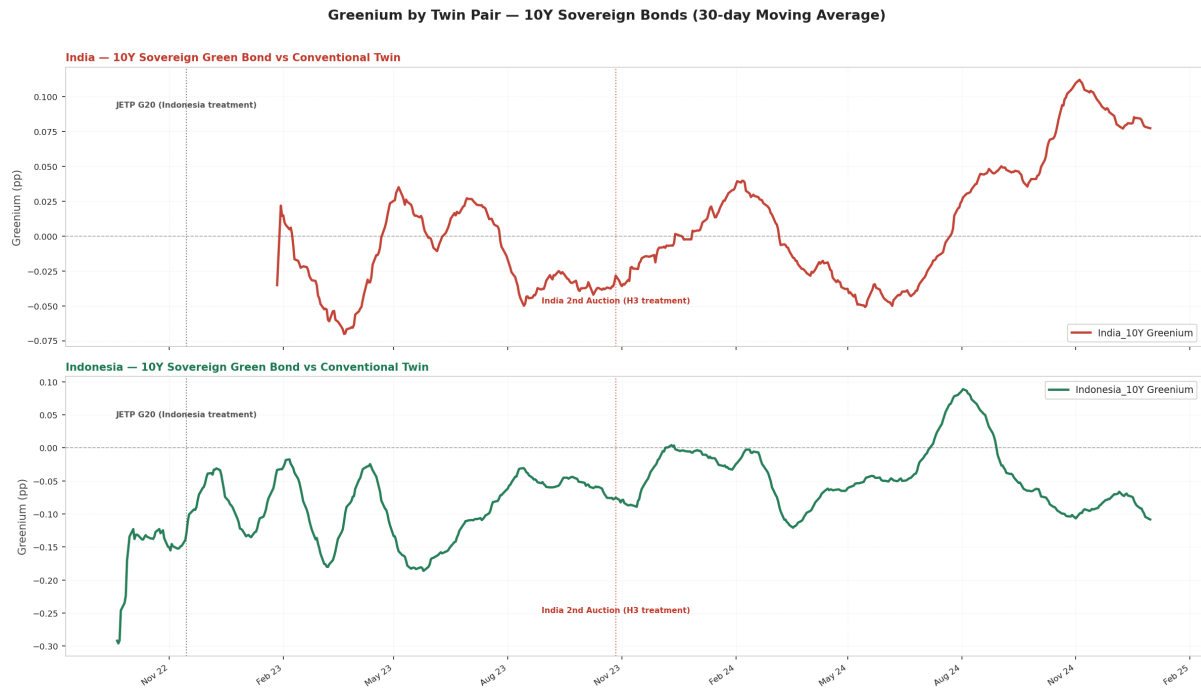


Figure B.1: Greenium by twin pair — 10Y sovereign bonds, India and Indonesia (30-day moving average). Vertical dashed lines mark the JETP G20 (15 Nov 2022) and India 2nd auction (27 Oct 2023) treatment dates. Source: Refinitiv Eikon; author's calculations.

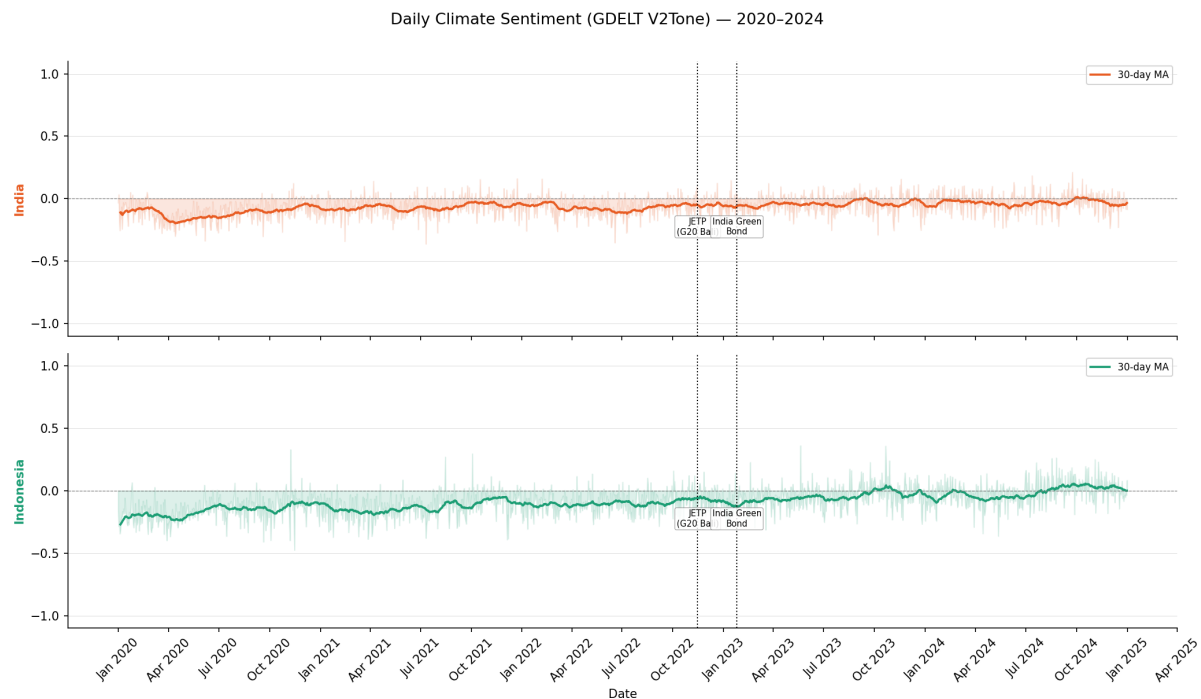


Figure B.2: Daily GDELT V2Tone climate sentiment — India and Indonesia, 2020–2024. The 30-day moving average (bold line) is flat throughout with no structural break at either treatment date, consistent with the H1 null result. Treatment dates marked by vertical dashed lines.

# Appendix C

## Full Regression Tables

Table C.1 presents the complete regression results for the primary Asian sample across all specifications.

Table C.1: Full Regression Results — Asia Panel

|                                  | M1                 | M2                    | M3                    | M3-India            | M3-Indonesia         | Stack A              |
|----------------------------------|--------------------|-----------------------|-----------------------|---------------------|----------------------|----------------------|
| <i>Panel A: Key coefficients</i> |                    |                       |                       |                     |                      |                      |
| Post Treatment (ATT)             |                    |                       | −0.0032<br>(0.0060)   | −0.0149<br>(0.0584) | −0.0311<br>(0.0331)  | −0.0284<br>(0.0337)  |
| $\Delta \log \text{VIX}$         |                    | −0.0035**<br>(0.0015) | −0.0036**<br>(0.0016) | −0.0019<br>(0.0018) | −0.0048*<br>(0.0027) | +0.0002<br>(0.0004)  |
| $\Delta \text{Sentiment}$        | 0.0002<br>(0.0002) | 0.0002<br>(0.0002)    | 0.0002<br>(0.0002)    | 0.0003<br>(0.0002)  | 0.0001<br>(0.0003)   | −0.0007<br>(0.0004)  |
| $\Delta \text{CDS Spread}$       |                    | −0.0001<br>(0.0003)   | −0.0001<br>(0.0003)   | −0.0002<br>(0.0004) | −0.0003<br>(0.0006)  | −0.0004*<br>(0.0002) |
| $\Delta \log \text{FX Rate}$     |                    | −0.0001<br>(0.0003)   | −0.0001<br>(0.0003)   | −0.0004<br>(0.0006) | −0.0001<br>(0.0004)  | —                    |
| <i>Panel B: Model fit</i>        |                    |                       |                       |                     |                      |                      |
| $N$ (obs)                        | 1,097              | 1,097                 | 1,097                 | 502                 | 595                  | 127                  |
| $R^2$                            | 0.0002             | 0.0183                | 0.0212                | 0.0252              | 0.0371               | 0.0922               |
| Pair FE                          | Yes                | Yes                   | Yes                   | Absorbed            | Absorbed             | Yes                  |
| Monthly FE                       | No                 | No                    | Yes                   | Yes                 | Yes                  | Yes                  |
| SE type                          | HC3                | HC3                   | HC3                   | HC3                 | HC3                  | HC3                  |

Notes: Dependent variable is  $\Delta \text{Greenium}_{i,t}$  in percentage points.  $\Delta \text{FX}$  omitted from Stack A. \*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ . HC3 SE in parentheses.

# Appendix D

## Standard Error Robustness

Table D.1 compares three standard error specifications.

Table D.1: Standard Error Robustness — Asia

| Variable           | M3 (pooled)           |                       | M5 (DiD, pooled)      |                       |                        |
|--------------------|-----------------------|-----------------------|-----------------------|-----------------------|------------------------|
|                    | HC3                   | PCSE                  | HC3                   | PCSE                  | Clustered ( $N = 2$ )  |
| $\Delta$ Sentiment | 0.0002<br>(0.0002)    | 0.0002<br>(0.0002)    | 0.0002<br>(0.0002)    | 0.0002<br>(0.0002)    | 0.0002<br>(0.0002)     |
| $\Delta$ logVIX    | −0.0035**<br>(0.0015) | −0.0035**<br>(0.0015) | −0.0036**<br>(0.0016) | −0.0036**<br>(0.0016) | −0.0036**<br>(0.0015)  |
| Post Treatment     |                       |                       | −0.0032<br>(0.0060)   | −0.0032<br>(0.0060)   | −0.0032***<br>(0.0009) |

Notes: HC3 = MacKinnon–White (1985). PCSE = Panel-Corrected SE via linearmodels with Bartlett kernel HAC. Clustered = pair-level ( $N = 2$ —severely downward biased; see text). \*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .



# Appendix E

## Placebo Tests

Table E.1 presents the placebo test results. All placebo Post coefficients are small and statistically insignificant, with  $p$ -values between 0.589 and 0.719.

Table E.1: Placebo Tests — Asia (M5 Pooled Specification)

| Specification        | Post $\hat{\beta}$ | SE       | $p$ -value | $N$   |
|----------------------|--------------------|----------|------------|-------|
| Placebo: −3 months   | −0.0009            | (0.0012) | 0.719      | 1,097 |
| Placebo: +3 months   | −0.0014            | (0.0021) | 0.622      | 1,097 |
| Placebo: +6 months   | −0.0010            | (0.0019) | 0.589      | 1,097 |
| Real treatment dates | −0.0032            | (0.0060) | 0.593      | 1,097 |

*Notes:* Each row re-estimates M5 with shifted treatment dates applied simultaneously to India and Indonesia. All other controls, FE, and SE assumptions identical to M5. HC3 SE in parentheses.

# Appendix F

## Parallel Trends Formal F-Tests

Table F.1 reports the joint Wald F-tests on all pre-event week dummies for the primary Asian sample, following [Roth \(2022\)](#).

Table F.1: Parallel Trends Joint F-Tests — Asia

| Country   | Pre-event weeks | F-statistic | df      | <i>p</i> -value | Result |
|-----------|-----------------|-------------|---------|-----------------|--------|
| India     | 9               | 1.006       | (9, 84) | 0.440           | Pass   |
| Indonesia | 7               | 0.740       | (7, 75) | 0.638           | Pass*  |

$H_0$ : all pre-event week dummies jointly equal to zero. Test follows [Roth \(2022\)](#). \* Indonesia pre-treatment window is  $\sim 55$  trading days; limited power.

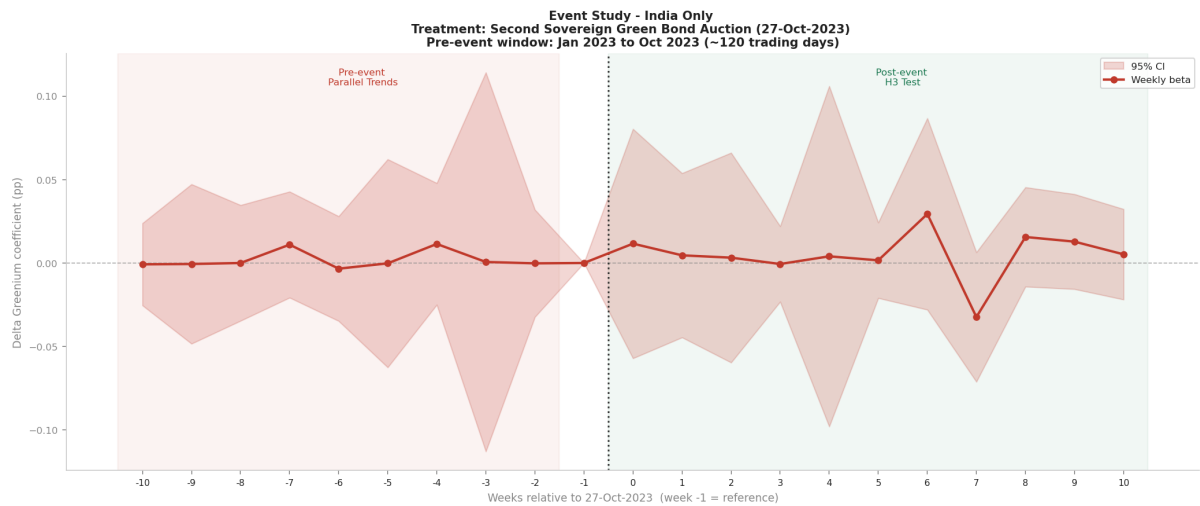


Figure F.1: Event study — India (treatment: 2nd green bond auction, 27 Oct 2023). Pre-event window: Jan–Oct 2023 ( $\sim 120$  trading days). Pre-event coefficients are jointly flat —  $F(9, 84) = 1.006$ ,  $p = 0.44$  — validating the DiD identification. No post-event shift detected (H3 null).

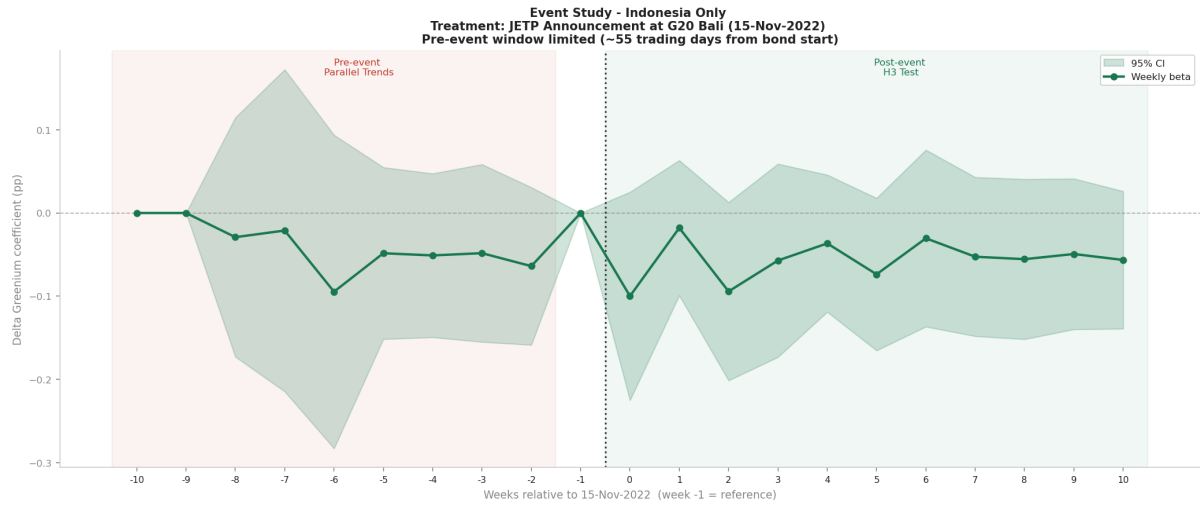


Figure F.2: Event study — Indonesia (treatment: JETP G20 Bali, 15 Nov 2022). Pre-event window limited to ~55 trading days from bond start. Weeks  $-10$  and  $-9$  are structural zeros (bond did not yet exist). Formal parallel trends test passes —  $F(7,75) = 0.740$ ,  $p = 0.64$  — but with limited statistical power.

# Appendix G

## European Robustness Results

Table G.1 presents the European DiD regression results. All Post Treatment coefficients are statistically indistinguishable from zero.

Table G.1: European Robustness Regression Results

|                      | <b>M5 Pooled</b>    | <b>M5-Germany</b>   | <b>M5-Denmark</b>   | <b>Stack A</b>      |
|----------------------|---------------------|---------------------|---------------------|---------------------|
| Post Treatment (ATT) | −0.0071<br>(0.0218) | −0.0009<br>(0.0008) | −0.0023<br>(0.0241) | −0.0010<br>(0.0008) |
| $\Delta \log VIX$    | −0.0000<br>(0.0003) | +0.0002<br>(0.0002) | −0.0004<br>(0.0006) | +0.0002<br>(0.0004) |
| $N$ (obs)            | 2,572               | 1,464               | 1,108               | 127                 |
| $R^2$                | 0.0020              | 0.0097              | 0.0032              | 0.0922              |
| Mean greenium (pp)   | −0.0224             | −0.0182             | −0.0280             | —                   |

Notes: HC3 SE in parentheses. Stack A uses Germany as treated and Denmark as control ( $\pm 180$  days around 2 Sep 2020). \*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

Table G.2 reports joint F-tests on post-event dummies for the European event study.

Table G.2: Post-Issuance Joint F-Tests — Europe

| <b>Country</b> | <b>Post-event weeks</b> | <b>F-statistic</b> | <b>df</b> | <b>p-value</b> | <b>Result</b> |
|----------------|-------------------------|--------------------|-----------|----------------|---------------|
| Germany        | 20                      | 0.603              | (20, 84)  | 0.900          | Null          |
| Denmark        | 20                      | 0.100              | (20, 87)  | 1.000          | Null          |

$H_0$ : all post-event week dummies jointly equal to zero. A null result confirms the greenium is stable after inaugural issuance.

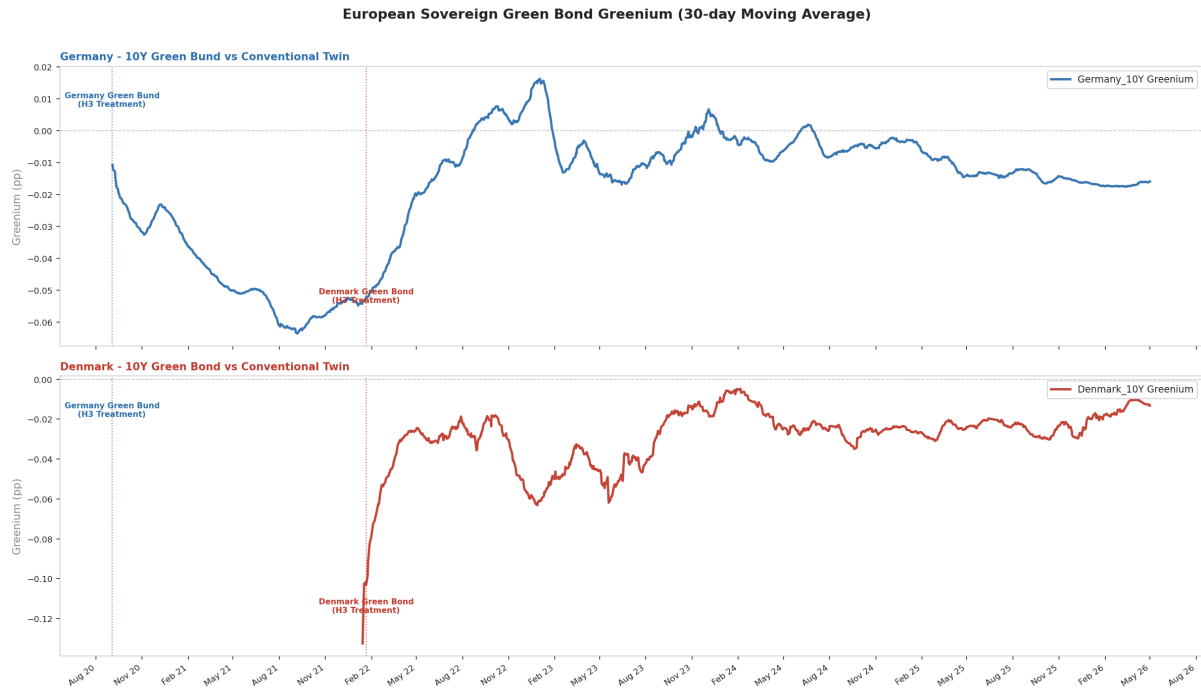


Figure G.1: European sovereign green bond greenium (30-day moving average) — Germany and Denmark. Both series show persistently negative, stable greeniums throughout the full sample. Vertical dashed lines mark inaugural issuance dates (Germany: 2 Sep 2020; Denmark: 21 Jan 2022). Source: Refinitiv Eikon; author's calculations.

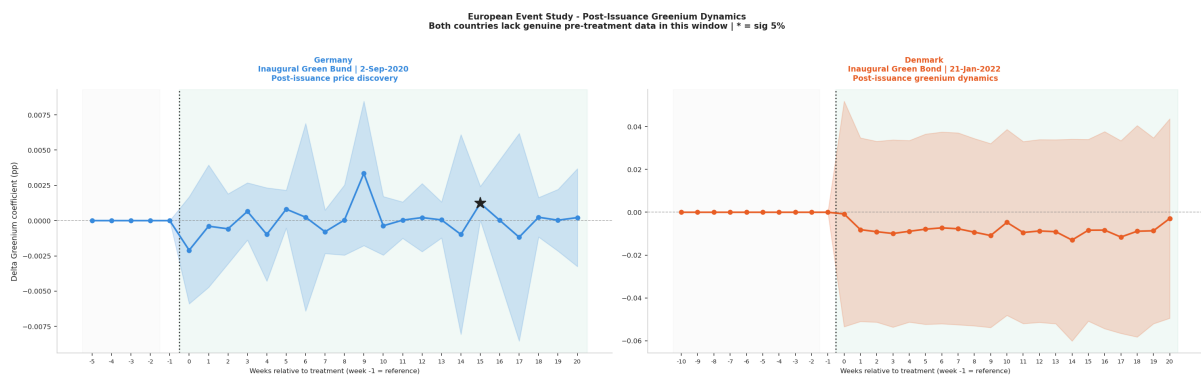


Figure G.2: Post-issuance greenium dynamics — Germany and Denmark (event study, side-by-side). Both countries show minor fluctuations around zero with no systematic post-issuance trend, consistent with a stable equilibrium greenium established at inaugural issuance.

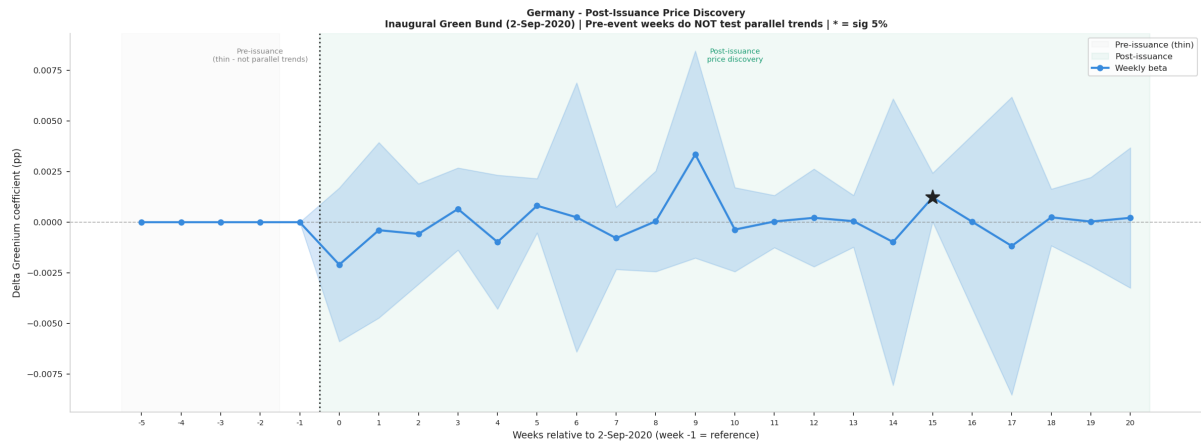


Figure G.3: Germany — post-issuance price discovery (inaugural Green Bund, 2 Sep 2020, 20-week window). Pre-issuance region (grey) does not test parallel trends. Post-issuance coefficients are jointly non-significant —  $F(20, 84) = 0.603$ ,  $p = 0.900$ .

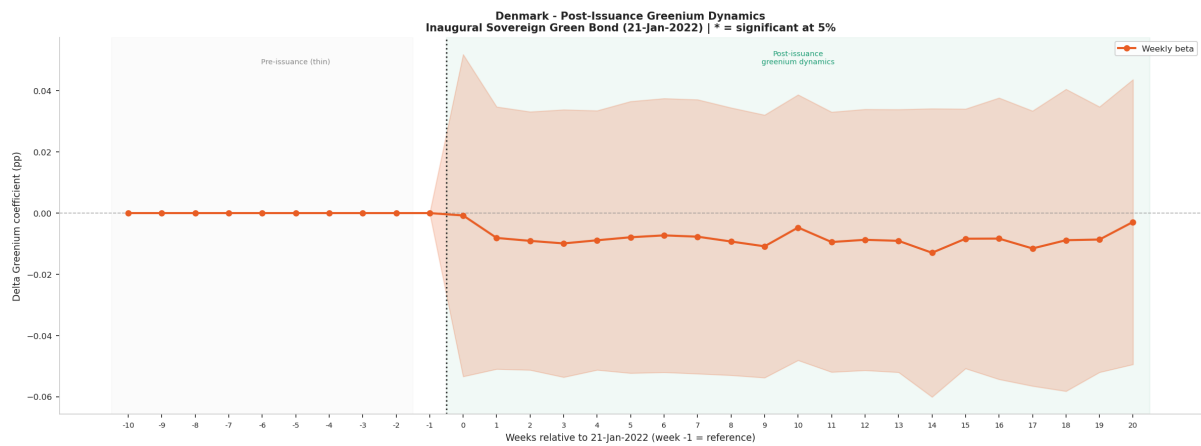


Figure G.4: Denmark — post-issuance greenium dynamics (inaugural sovereign green bond, 21 Jan 2022, 20-week window). Pre-issuance region is thin. Post-issuance coefficients are jointly non-significant —  $F(20, 87) = 0.100$ ,  $p = 1.000$ .

# Appendix H

## Stacked DiD Verification

### Stack A verification:

- Window: 18 May 2022 to 15 May 2023 ( $\pm 180$  days around 15 Nov 2022)
- Countries: Indonesia (treated), India (clean control)
- Indonesia Post = 1 from 15 Nov 2022: confirmed
- India Post = 0 throughout: confirmed (not yet treated)
- Total observations: 127 | Valid DiD: **Yes**

### Stack B verification:

- Window:  $\approx \pm 180$  days around 27 October 2023
- Countries: India only (Indonesia already treated — excluded)
- Total observations: 257 | Valid DiD: **No — pre-post only**

# **Appendix I**

## **Granger Causality Tests**

Table I.1 reports Granger causality tests for the relationship between signalling noise and the greenium, run separately for India and Indonesia. Maximum lags tested: 5 trading days.



Table I.1: Granger Causality Tests — Signalling Noise and Greenium

| Lag                                                                 | F-statistic | p-value | Sig. | Verdict            |
|---------------------------------------------------------------------|-------------|---------|------|--------------------|
| <i>Panel A: India (N = 502) — Direction 1: Noise → Greenium</i>     |             |         |      |                    |
| 1                                                                   | 0.418       | 0.518   | n.s. | $H_0$ not rejected |
| 2                                                                   | 0.982       | 0.375   | n.s. | $H_0$ not rejected |
| 3                                                                   | 0.469       | 0.704   | n.s. | $H_0$ not rejected |
| 4                                                                   | 0.423       | 0.792   | n.s. | $H_0$ not rejected |
| 5                                                                   | 0.441       | 0.820   | n.s. | $H_0$ not rejected |
| <i>Panel B: India (N = 502) — Direction 2: Greenium → Noise</i>     |             |         |      |                    |
| 1                                                                   | 0.669       | 0.414   | n.s. | $H_0$ not rejected |
| 2                                                                   | 5.450       | 0.005   | ***  | $H_0$ rejected     |
| 3                                                                   | 3.263       | 0.021   | **   | $H_0$ rejected     |
| 4                                                                   | 2.560       | 0.038   | **   | $H_0$ rejected     |
| 5                                                                   | 1.991       | 0.079   | *    | $H_0$ rejected     |
| <i>Panel C: Indonesia (N = 595) — Direction 1: Noise → Greenium</i> |             |         |      |                    |
| 1                                                                   | 1.007       | 0.316   | n.s. | $H_0$ not rejected |
| 2                                                                   | 0.505       | 0.604   | n.s. | $H_0$ not rejected |
| 3                                                                   | 0.340       | 0.797   | n.s. | $H_0$ not rejected |
| 4                                                                   | 0.538       | 0.708   | n.s. | $H_0$ not rejected |
| 5                                                                   | 0.481       | 0.790   | n.s. | $H_0$ not rejected |
| <i>Panel D: Indonesia (N = 595) — Direction 2: Greenium → Noise</i> |             |         |      |                    |
| 1                                                                   | 1.031       | 0.310   | n.s. | $H_0$ not rejected |
| 2                                                                   | 1.088       | 0.338   | n.s. | $H_0$ not rejected |
| 3                                                                   | 0.853       | 0.465   | n.s. | $H_0$ not rejected |
| 4                                                                   | 0.708       | 0.587   | n.s. | $H_0$ not rejected |
| 5                                                                   | 0.547       | 0.741   | n.s. | $H_0$ not rejected |

$H_0$ :  $X$  does not Granger-cause  $Y$ . F-statistic from SSR-based F-test. \*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ . India Direction 2 (Greenium → Noise): significant at lags 2–4, consistent with reverse causality where market pricing influences media tone coverage. Overall verdict: H2 null confirmed.